

Workshop 1

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Exam exercise for Module 1: Wind speed distributions

In this workshop we consider a continuous probability distribution called the Weibull distribution. Among other things, it is used to model wind speed distributions.

Part I is a purely theoretical exercise which you should answer using pen and paper only. For Part II and III we recommend that you answer the exercises using Rmarkdown (you can simply use the exam Rmarkdown file as a starting point).

There are many calculations of integrals in this exercise. At the exam, you will not have the time to go through all of them in detail. It is enough to go through the main points.

Part I: The Weibull distribution

The Weibull distribution depends on two parameters $k > 0$ and $\lambda > 0$. If X follows a Weibull distribution with parameters k and λ , we write $X \sim \text{weibull}(k, \lambda)$. In this case, X has the probability density function

$$f(x) = \begin{cases} \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

The parameter k is called the shape parameter, since it determines the shape of the distribution, while λ is called the scale parameter, because it works by scaling the x -axis. The plots below show the density of a Weibull distribution with $\lambda = 1$ for different values of k .

1. Suppose $X \sim \text{weibull}(k, \lambda)$. Show that the distribution function of X is $F(x) = 1 - e^{-(x/\lambda)^k}$ for $x > 0$. (Hint: Recall that the distribution function is defined by $F(x) = P(X \leq x)$. Write this as an integral and apply the substitution $u = (x/\lambda)^k$.)

$$f(x) = \begin{cases} \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

$$F(x) = P(X \leq x) = \int_0^x f(x) dx = \int_0^x \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k} dx$$

$$u = \left(\frac{x}{\lambda}\right)^k \rightarrow \frac{du}{dx} = \frac{k}{\lambda^k} \cdot x^{k-1} \rightarrow du = \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} dx$$

$$F(x) = \int_0^u e^{-u} du = [-e^{-u}]_0^u = [-e^{-(x/\lambda)^k}]_0^{(x/\lambda)^k} = -e^{-(x/\lambda)^k} + e^0 = 1 - e^{-(x/\lambda)^k}.$$

2. Show that the distribution function $F(x)$ satisfies

$$\ln(-\ln(1 - F(x))) = -k \ln(\lambda) + k \ln(x).$$

$$\begin{aligned}
\ln(-\ln(1 - F(x))) &= \ln(-\ln(1 - (1 - e^{-(x/\lambda)^k}))) \\
&= \ln(-\ln(e^{-(x/\lambda)^k})) = \ln(-(x/\lambda)^k) \\
&= \ln((x/\lambda)^k) = k \cdot \ln(x/\lambda) = k \ln(x) - k \ln(\lambda) \\
&= -k \ln(\lambda) + k \ln(x)
\end{aligned}$$

3. Show that the mean value of X is $\lambda\Gamma(1 + \frac{1}{k})$. (Hint: Remember that the gamma function is given by $\Gamma(s) = \int_0^\infty u^{s-1}e^{-u}du$.)

$$\begin{aligned}
\mu &= \int_{-\infty}^{\infty} x \cdot f(x) dx, \\
f(x) &= \begin{cases} \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}, & x > 0, \\ 0, & x \leq 0, \end{cases} \\
\mu &= \int_0^\infty x \cdot \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k} dx \\
u &= \left(\frac{x}{\lambda}\right)^k, \quad du = \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} dx \\
\mu &= \int_0^\infty x \cdot e^{-u} du, \quad x = \lambda u^{-1/k} \\
\mu &= \int_0^\infty \lambda u^{-1/k} \cdot e^{-u} du \\
\Gamma(s) &= \int_0^\infty u^{s-1} \cdot e^{-u} du \\
1/k &= s - 1 \rightarrow s = 1/k + 1 \\
\therefore \mu &= \lambda\Gamma(1/k + 1)
\end{aligned}$$

4. By similar calculations, one may show that the variance of X is $\lambda^2(\Gamma(1 + \frac{2}{k}) - \Gamma(1 + \frac{1}{k})^2)$. What is the standard deviation?

$$\sigma = \lambda \sqrt{\Gamma(1 + \frac{2}{k}) - \Gamma(1 + \frac{1}{k})^2}$$

Part II: Wind speed measurements

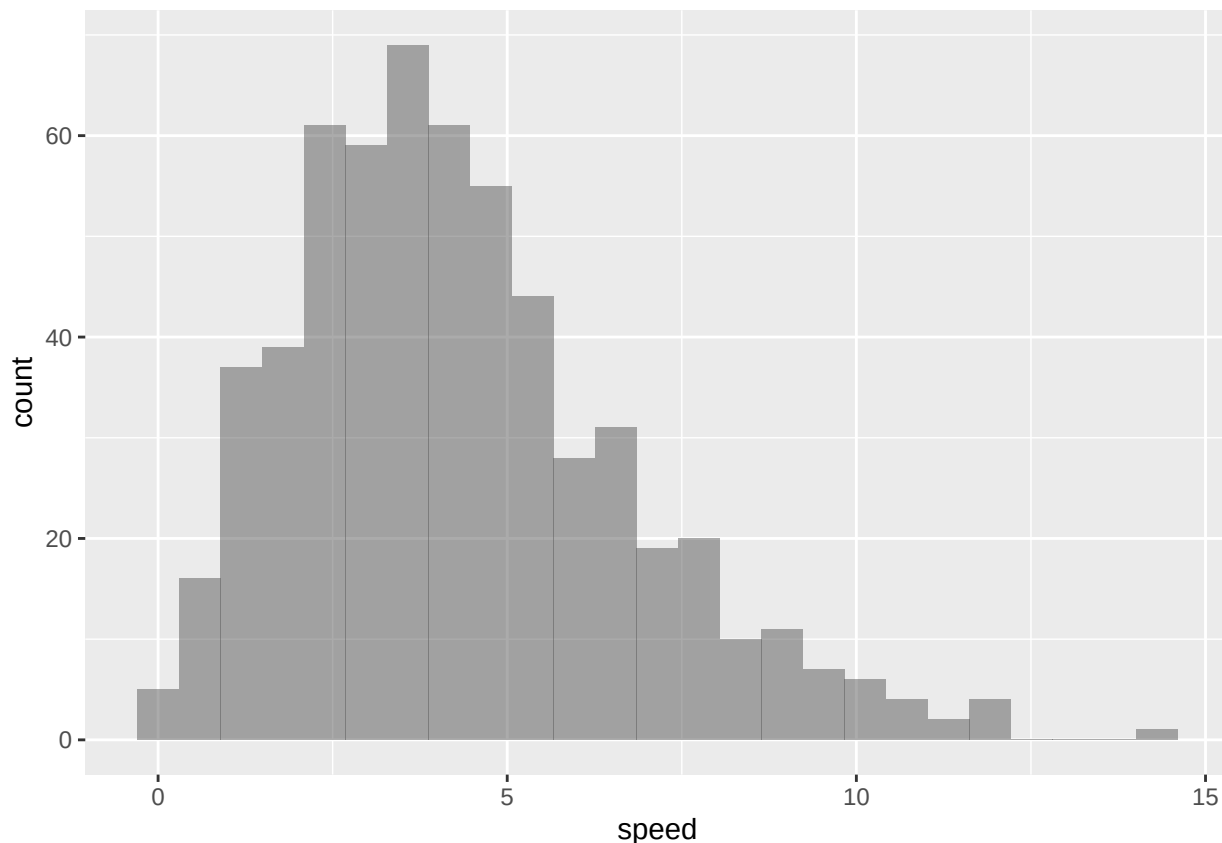
In this part we consider a data set containing wind speed measurements from a Danish weather station located at Sjølsmark. The data set contains the wind speed measured at 12 noon every day of January in the years 2001-2019. We first load the data set:

```
speed<-read.delim("https://asta.math.aau.dk/datasets?file=windSpeed.txt",header=FALSE)[,1]
```

1. Draw a histogram of the wind speed observations by editing the R chunk below. Explain how a histogram is constructed. Do you think the observations come from a normal distribution?

```
library(mosaic)
```

```
gf_histogram(~ speed, bins=25)
```



In the following we will convince ourselves that the data actually comes from a Weibull distribution. We order the $n = 589$ observations from smallest to largest

$$x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}.$$

```
speed_sorted <- sort(speed)
```

2. Argue that $F(x_{(i)}) \approx \frac{i}{n}$ for $i = 1, \dots, n$. (Hint: How many observations are less than or equal to $x_{(i)}$?)

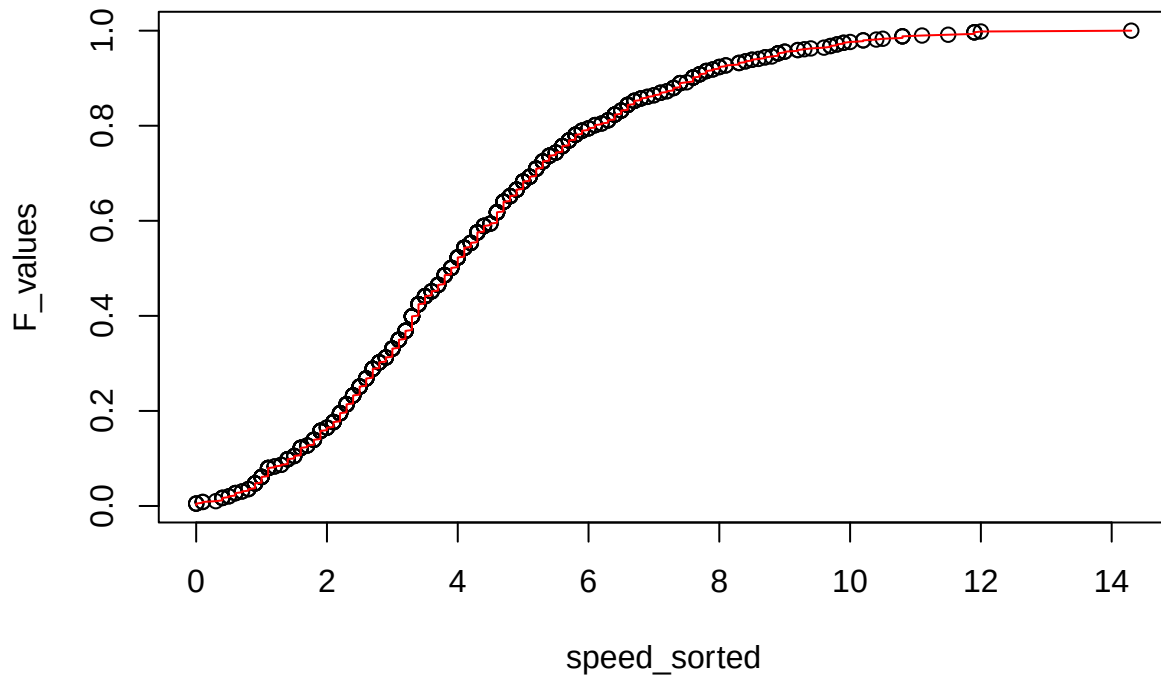
```
# Number of observations
n <- length(speed_sorted)

# Create the index i = 1, 2, ..., n
i <- 1:n

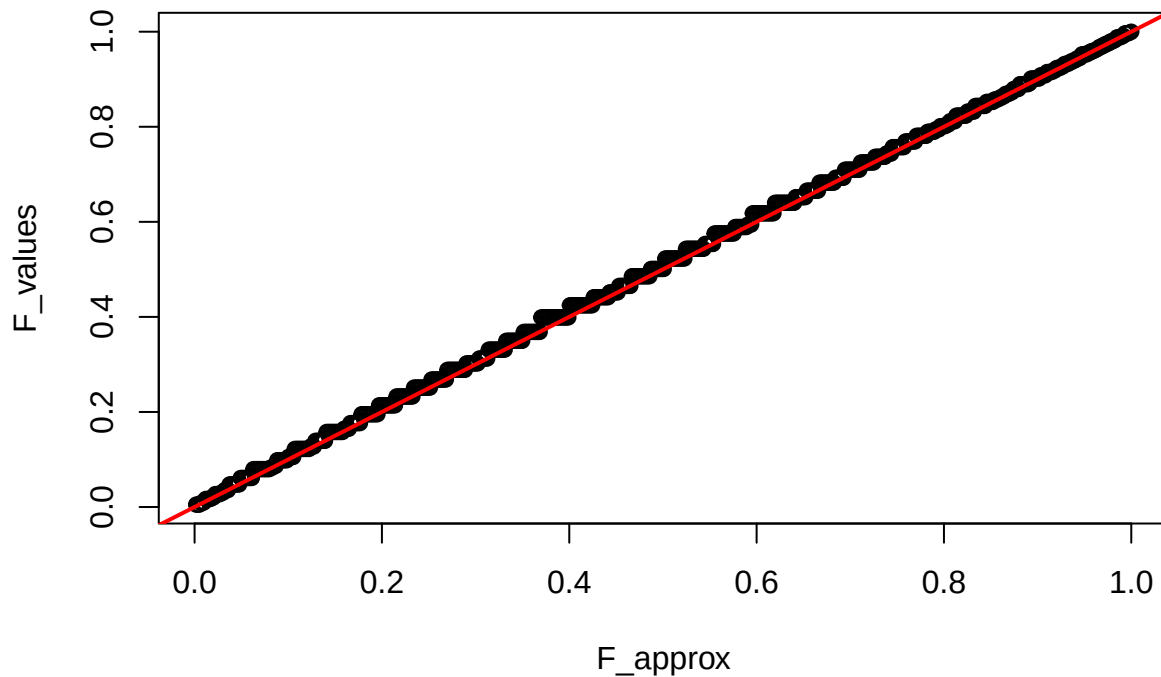
# Approximate CDF at x_(i) = i/n
F_approx <- i / n

# speed_sorted is already your sorted data, then Evaluate CDF at each x_(i)
F_empirical <- ecdf(speed_sorted) # creates the ECDF function
F_values <- F_empirical(speed_sorted)

plot(speed_sorted, F_values)
lines(speed_sorted, F_approx, col="red")
```



```
# Combine into a plot to see. If it's close to the y=x plot it means that they are close
plot(F_approx, F_values)
abline(0, 1, col="red", lwd=2)
```



3. Using Exercise 2 in Part I, argue that if the observations come from a `weibull( $k, \lambda$ )` distribution, then

$$\ln(-\ln(1 - \frac{i}{n})) \approx -k \ln(\lambda) + k \ln(x_{(i)}).$$

If it's a Weibull distribution, as shown per the previous analysis, the following expression is true when $f(x) = \text{weibull}(x, k, \lambda)$:

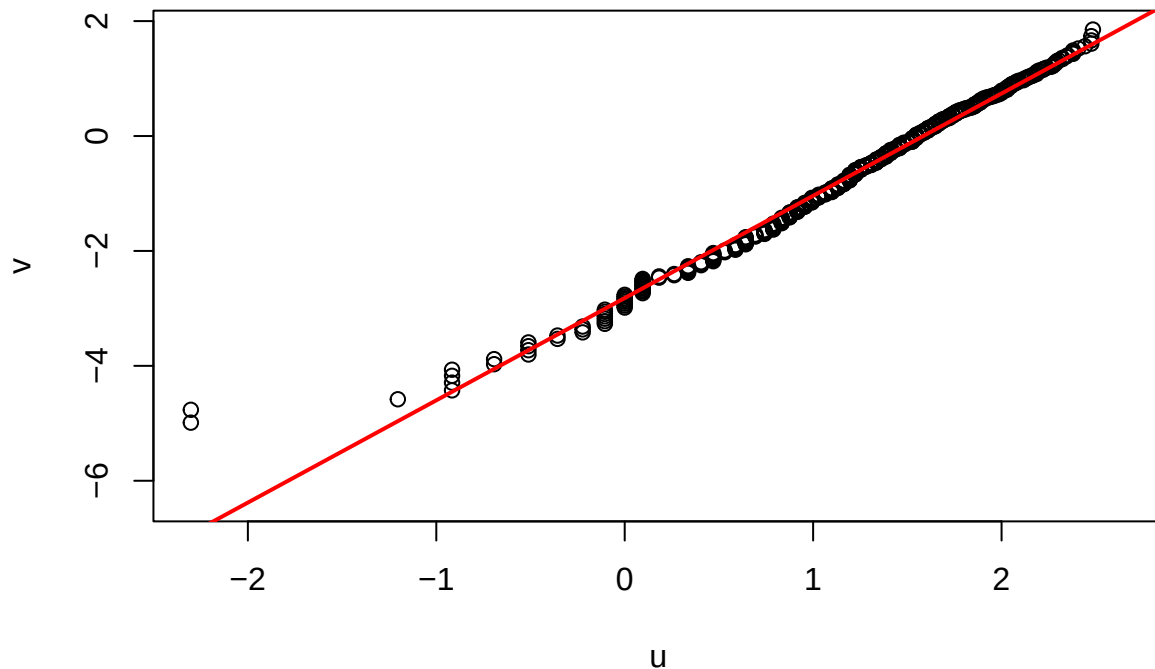
$$\ln(-\ln(1 - F(x))) = -k \ln(\lambda) + k \ln(x)$$

Now we know that $F(x) \approx \frac{i}{n}$, then:

$$\ln(-\ln(1 - \frac{i}{n})) \approx -k \ln(\lambda) + k \ln(x_{(i)})$$

The code below computes a vector containing the values $v_i = \ln(-\ln(1 - \frac{i}{n}))$ and a vector containing the values $u_i = \ln(x_{(i)})$.

```
n<-length(speed)
sortedSpeed<-sort(speed)
u<-log(sortedSpeed)
CDF<-(1:n)/n
v<-log(-log(1-CDF))
plot(u,v)
abline(-2.82, 1.78, col="red", lwd=2)
```



4. Argue that the points (u_i, v_i) should lie approximately on a straight line if the observations come from a `weibull(k, λ)` distribution. Edit the code above to check that this is the case.

The term $-k \ln(\lambda) + k \ln(x_{(i)})$ can be thought as a linear function of $\ln(x_{(i)})$, where $-k \ln(\lambda)$ is a constant and k is the slope. If we then plot $\ln(x_{(i)})$ against $\ln(-\ln(1 - \frac{i}{n}))$ the resulting intercept would equal to $-k \ln(\lambda)$, and the slope would just be k .

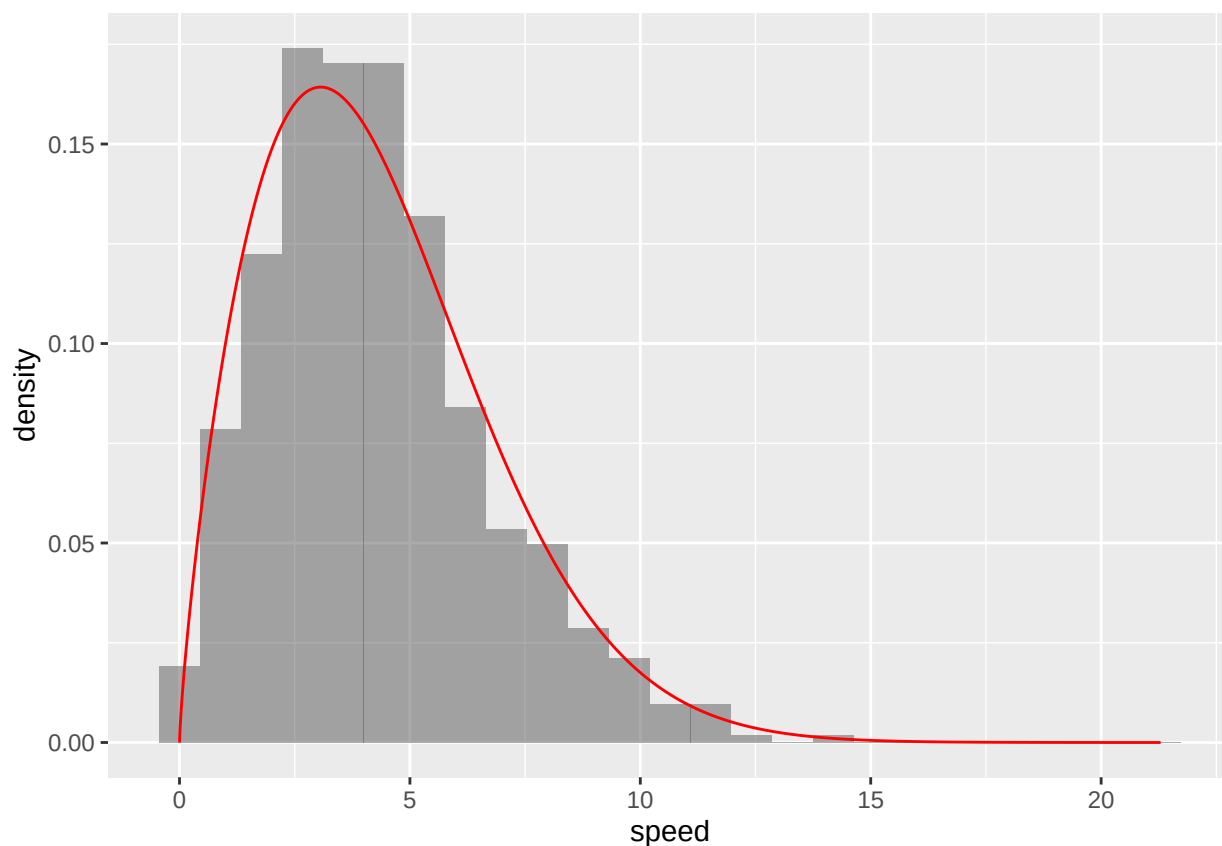
5. The intercept and slope of the line can be found to be -2.82 and 1.78 , respectively. Use this to give estimates of the parameters k and λ of the model. Insert these values in the code below to plot the histogram together with the approximate density (`shape` is k and `scale` is λ).
6. Slope is directly the value of k

$$k = 1.78$$

7. λ can be calculated from the intercept

$$\begin{aligned} -k \cdot \ln(\lambda) &= -2.82 \\ \lambda &= e^{-2.82/(-k)} = e^{-2.82/(-1.78)} = 4.87 \end{aligned}$$

```
k <- 1.78
lambda <- 4.87
gf_dhistogram( ~ speed, bins = 25) %>%
  gf_dist("weibull", shape = k, scale = lambda, col = "red")
```



Part III: Sample mean and the central limit theorem

In this last exercise, we investigate the distribution of the sample mean when a random sample is taken from a population having a $\text{weibull}(k, \lambda)$ distribution. We will use the values of k and λ that you found in Part II, Exercise 5 to mimic a sample of wind speed measurements.

1. What is the mean and standard deviation in the population distribution? Use the calculations from Part I. (Hint: You can use the function `gamma()` in R to compute the gamma function.)

The mean value of X is $\lambda\Gamma(1 + \frac{1}{k})$ and the standard deviation is

$$\sigma = \lambda \sqrt{\Gamma(1 + \frac{2}{k}) - \Gamma(1 + \frac{1}{k})^2}$$

```
mu <- lambda*gamma(1+1/k)
sigma <- lambda*sqrt(gamma(2+1/k) - gamma(1+1/k)^2)
```

```
mu
```

```
## [1] 4.333259
```

```
sigma
```

```
## [1] 3.765825
```

2. Suppose that a sample consists of 30 observations from this distribution. We denote the sample mean by $\bar{x}$. Using the central limit theorem, answer the following questions:

- What is the expected value of $\bar{x}$? CLT explains that the expected value of $\bar{x}$ is the mean of the distribution, calculated before as:

```
x_bar <- mu
x_bar
```

```
## [1] 4.333259
```

- What is the standard deviation of $\bar{x}$ (also called the standard error)?

```
n <- 30
se <- sigma/sqrt(n)
se
```

```
## [1] 0.6875425
```

- What is the approximate distribution of $\bar{x}$?

By the Central Limit Theorem, for sufficiently large n , the sample mean $\bar{x}$ is approximately normally distributed, regardless of the shape of the original Weibull distribution.

$$\bar{x} \approx N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

The code below generates 30 independent realizations of a Weibull distribution with parameters k and λ . One may think of this of as simulated random sample of 30 independent wind speed observations.

```
x<-rweibull(30, shape=k, scale=lambda)
mean(x)
```

```
## [1] 3.634376
```

4. Insert the values of k and λ from Part II, Exercise 5 in the code. Run the command a few times. Is each sample mean close to what you expected?

Use `replicate` to repeat the sampling 500 times and save each mean value in the vector $\bar{x}$:

```
x_bar <- replicate(500, mean(rweibull(30, shape=k, scale = lambda) ))
x_bar
```

```
## [1] 3.845222 3.651171 4.242949 4.176083 4.540233 3.714912 4.264250 4.371786
## [9] 4.976349 4.540853 4.603915 3.756618 4.312126 4.190347 4.030274 3.828996
## [17] 4.072956 3.821481 4.611300 4.630416 3.892251 3.564336 4.675768 3.998082
## [25] 4.122458 4.223864 4.114054 3.193510 4.777374 3.551902 5.877124 3.933684
## [33] 3.781694 3.847600 4.722122 4.908929 4.721134 5.032017 4.028236 5.233750
## [41] 3.773327 4.055669 4.411725 4.952524 4.461126 3.933353 3.648833 4.816650
## [49] 4.312470 4.187674 4.209227 4.561831 4.860648 4.158890 4.366965 3.806009
## [57] 4.254700 4.846759 4.800986 4.177788 3.938355 4.490311 4.361856 3.764619
## [65] 4.678366 4.575375 4.469934 3.482356 4.158362 3.955167 4.592690 4.266690
## [73] 3.738555 4.164834 4.722558 4.094585 4.747283 4.476818 3.887272 3.357189
## [81] 4.363377 3.605366 3.700932 4.475937 3.644412 4.384817 4.158772 4.570477
## [89] 4.590670 3.907824 3.698587 4.257977 3.458221 5.003088 4.282102 4.867931
## [97] 3.877662 4.635316 5.732865 4.692291 3.919142 4.396253 4.555609 4.470508
## [105] 4.631229 4.766553 4.301101 4.157756 4.046403 3.778945 4.520279 3.725604
## [113] 4.594422 4.774602 4.656838 4.918932 4.024182 4.602436 3.801721 4.360824
## [121] 3.689069 4.743544 4.158655 3.908794 3.902589 4.369394 4.540726 3.173052
## [129] 4.399635 4.378879 4.983199 5.461556 5.060709 4.154456 4.873816 4.765226
```

```
## [137] 3.933636 4.635477 4.203176 4.613842 4.198530 4.038890 3.877556 4.611113
## [145] 4.092704 4.772021 3.457132 3.566041 4.496347 4.692031 4.482575 4.505666
## [153] 4.208392 4.315779 4.586789 4.948470 3.949018 5.303099 3.746556 3.420401
## [161] 4.667496 4.148812 5.343590 4.773197 4.206718 4.480535 4.393945 5.077425
## [169] 4.027889 3.801394 3.897178 3.390995 4.878441 3.859947 3.849241 3.948007
## [177] 4.379562 4.073632 4.406070 4.579483 4.173918 4.503219 4.785225 4.393746
## [185] 4.271303 4.044752 4.555463 5.132300 5.264547 4.689905 4.347427 4.142593
## [193] 4.901429 3.805484 3.473075 4.475844 4.393731 4.145492 5.220587 4.919542
## [201] 4.926213 4.624933 4.160199 4.053593 3.907658 4.230510 4.293513 4.584490
## [209] 3.929744 5.182341 4.340219 4.887730 3.906198 4.460340 4.059665 4.383268
## [217] 4.663271 3.972561 5.273068 4.269493 4.153128 4.495352 4.739002 4.831056
## [225] 4.699913 4.723770 3.883454 4.273729 3.596506 4.261354 4.102443 4.491551
## [233] 5.169614 3.875860 4.922524 4.052322 4.388014 3.694272 3.922311 4.335580
## [241] 4.785156 3.962224 3.952757 4.867381 3.866276 4.270232 5.040839 3.920845
## [249] 3.942117 4.309048 4.611345 4.850765 4.406932 4.238154 4.742384 3.925031
## [257] 4.927107 4.134859 3.636689 5.016580 3.819048 4.942447 4.319463 4.714662
## [265] 4.037703 4.484629 4.789799 4.389838 4.584645 4.293936 3.869206 5.051525
## [273] 5.002377 4.165874 4.041784 4.980235 4.508868 4.694122 4.609249 5.018138
## [281] 3.923398 5.257941 3.997441 3.764895 4.043648 4.449850 4.736175 4.541919
## [289] 3.943726 4.281094 4.088802 4.076116 4.937184 3.981618 3.539172 3.745084
## [297] 4.068136 4.784898 5.080916 4.990239 3.181925 4.614664 4.570886 4.425087
## [305] 4.136826 4.464554 3.958941 4.246319 4.554327 3.604091 4.421189 4.397714
## [313] 3.627882 4.551837 4.497271 4.343881 4.469807 4.379368 4.218786 4.160354
## [321] 4.049173 4.642866 5.049614 4.359678 4.990072 3.939056 4.670117 4.423483
## [329] 4.832982 3.212638 4.506281 4.008648 4.974894 4.526625 4.709758 4.035136
## [337] 4.473438 4.651986 4.418528 3.472422 4.769049 4.126938 4.217370 3.893277
## [345] 4.949684 5.016466 4.886295 4.279144 4.984072 4.982137 4.117244 4.966000
## [353] 5.543035 3.935003 5.318131 4.536694 5.114031 4.744376 5.270723 4.181048
## [361] 4.074902 3.753684 4.060132 4.352405 4.669440 4.388082 3.944630 3.454274
## [369] 4.676160 4.968504 4.514601 4.883879 5.450528 4.116890 4.236061 4.119966
## [377] 3.947985 3.723581 4.425549 3.847828 4.458875 4.882580 4.285868 4.437430
## [385] 3.672339 4.437990 5.808821 3.871410 4.982457 4.425476 4.566930 3.776499
## [393] 5.347651 3.948872 4.409084 4.485623 4.873885 4.759169 4.312978 3.828608
## [401] 3.848216 3.306134 4.220271 4.228181 3.452420 5.032677 4.587248 4.114129
## [409] 3.581296 4.023999 4.751163 3.761245 4.801312 3.910384 3.310246 3.668704
## [417] 3.734021 4.180404 4.442873 4.486750 4.973841 3.702034 5.386561 3.726091
## [425] 4.111629 4.223493 3.949935 4.310517 4.147636 4.491224 3.032002 4.751845
## [433] 4.236337 3.309521 4.204810 3.671495 4.476516 5.179963 4.611519 4.056279
## [441] 4.510572 4.380032 3.993139 3.586627 4.019011 3.942566 4.830933 4.578988
## [449] 5.647572 4.966002 4.729317 4.718468 4.015206 4.462966 3.954474 4.214030
## [457] 4.020957 3.965553 5.035305 3.685208 4.663257 4.849810 4.074824 3.860396
## [465] 4.726138 3.987068 4.344055 4.350070 4.357547 4.670308 3.993341 3.640282
## [473] 4.125596 4.092387 4.244156 4.480816 4.530723 4.297097 4.507229 4.439160
## [481] 4.399385 3.619011 4.876698 4.598467 4.411435 4.594313 3.931925 4.391333
## [489] 3.556437 3.647892 3.647137 3.930472 4.830687 4.126645 4.207763 3.747908
## [497] 4.264790 4.556513 4.023975 4.430898
```

The values inside `x_bar` are somewhat close to the calculated value of μ .

5. Calculate the mean and standard deviation of the values in `x_bar`. How do they match with what you expected?

```
mean(x_bar)
```

```
## [1] 4.333197
```



```
sd(x_bar)
```

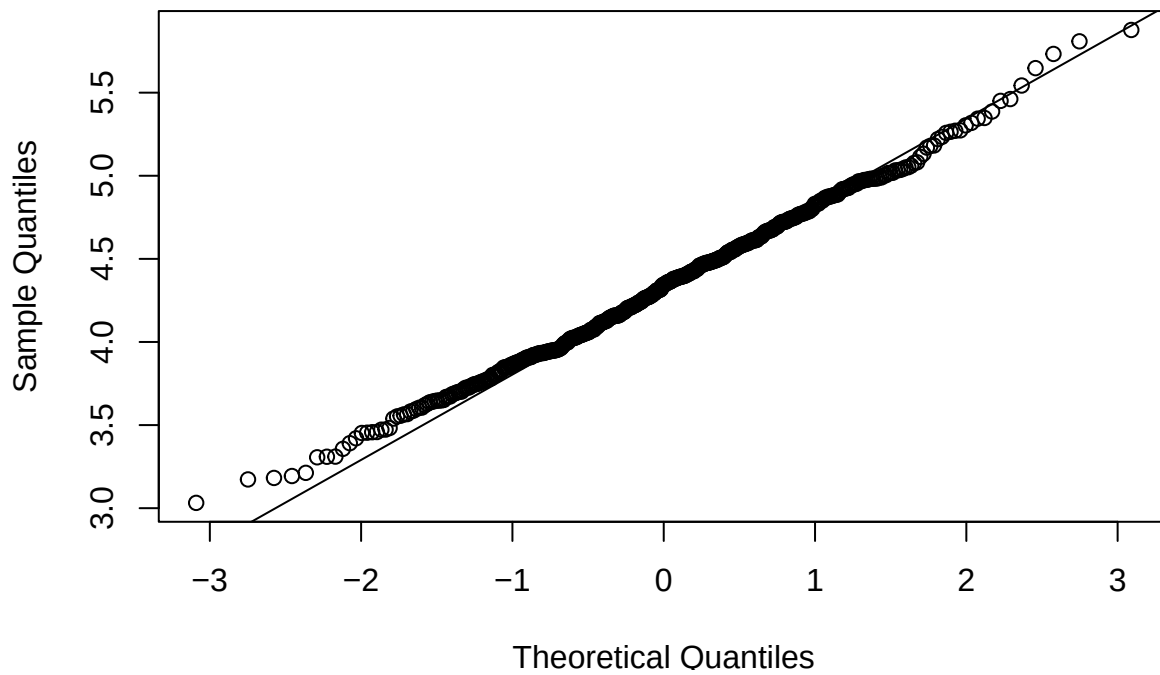
```
## [1] 0.476875
```

Both values align with the previously calculated `mu` and `sigma`.

6. Make a QQ-plot to assess the distribution of `x_bar`. Does this look like what you would expect?

```
qqnorm(x_bar) # plots the quantiles of x_bar against the theoretical quantiles of a standard normal dis  
qqline(x_bar) # adds a straight line through the first and third quartiles, if the points fall roughly
```

Normal Q-Q Plot



Exam exercise: Seasonal wind speed

It is highly recommended that you answer the exam using Rmarkdown (you can simply use the exam Rmarkdown file as a starting point).

The data set for this exam problem contains measurements of the quarterly average wind speed in the years 2001-2019. The measurements were made at the Danish weather station Sjølsmark in the years 2001-2019.

We first read in the data:

```
library(mosaic)
wind <- read.delim("https://asta.math.aau.dk/datasets?file=speed_quarterly.txt", sep="")
head(wind)
```

```
##   quarter   speed
## 1  winter 3.571196
## 2  winter 4.448287
## 3  winter 3.590417
## 4  winter 3.306593
## 5  winter 4.047176
## 6  winter 3.395046
```

We saw in Exam exercise 1 that wind speed measurements are Weibull distributed. However, because of the central limit theorem, when we take the average of hourly measurements from three months, these values will be close to having a normal distribution. Hence, we will assume in this workshop that the quarterly averages are normally distributed.

Throughout the exercise, we use a significance level of $\alpha = 0.05$.

Part I: Comparison of wind speed for spring and autumn

In part I we compare the wind speed in spring and autumn. The questions for Part I should be answered using pen and paper. You may use R as a calculator. You can use the output from `favstats` in your computations:

```
favstats(~speed|quarter, data=wind)
```

```
##   quarter   min      Q1  median      Q3     max    mean      sd  n
## 1  autumn 2.397207 3.010508 3.097482 3.230105 3.627427 3.094134 0.2532494 19
## 2  spring 2.877559 3.117040 3.380616 3.572803 3.794384 3.346272 0.2772962 19
## 3  summer 2.246513 2.699706 2.848324 3.059420 3.438134 2.878752 0.2751879 19
## 4  winter 2.940046 3.382014 3.805370 4.069144 4.448287 3.721420 0.4084774 19
##   missing
## 1         0
## 2         0
## 3         0
## 4         0
```

1. Make an F -test to check whether the variance is the same in spring and autumn.

$$H_0 : \sigma_{spring}^2 = \sigma_{autumn}^2 \quad H_A : \sigma_{spring}^2 \neq \sigma_{autumn}^2$$

This is a two-sided test because we care about whether spring's variance is higher or lower than autumn's.

```
# SDs
sd_autumn <- favstats(~speed|quarter, data=wind[wind$quarter == "autumn", ])$sd
sd_spring <- favstats(~speed|quarter, data=wind[wind$quarter == "spring", ])$sd

# Degrees of freedom
```

```

n_autumn <- favstats(~speed|quarter,data=wind[wind$quarter == "autumn", ])$n
n_spring <- favstats(~speed|quarter,data=wind[wind$quarter == "spring", ])$n
df_autumn <- n_autumn-1
df_spring <- n_spring-1

# F-statistic
Ft <- sd_spring^2/sd_autumn^2

# Two-sided p-value
p_value_1 <- 2 * min(pf(Ft, df1=df_spring, df2=df_autumn),
                     1 - pf(Ft, df1=df_spring, df2=df_autumn))
p_value_1

```

```
## [1] 0.7044691
```

The p-value is large, so we do not reject the null hypothesis of equal variances. We can assume the variances in spring and autumn are approximately equal.

2. Make a t-test to compare the mean wind speed in spring and autumn. What is the null hypothesis and the corresponding alternative hypothesis? Explain how the test statistic and p-value are computed? What is the conclusion of the analysis? $H_0 : \mu_{spring} = \mu_{autumn}$ $H_A : \mu_{spring} \neq \mu_{autumn}$

This is a two-sided test because we care about whether spring's mean is higher or lower than autumn's.

Because we assume equal variances, we combine the variances from both groups:

```

df<-n_autumn+n_spring-2
sp_2 <- ((n_autumn-1)*sd_autumn^2+(n_spring-1)*sd_spring^2)/(df)
sp <- sqrt(sp_2)

```

Compute the SE of the difference:

```
SE <- sp*sqrt(1/n_autumn + 1/n_spring)
```

How far apart are the sample means compared to the natural variability of the data? Compute the t-statistic:

```

mu_autumn <- favstats(~speed|quarter,data=wind[wind$quarter == "autumn", ])$mean
mu_spring <- favstats(~speed|quarter,data=wind[wind$quarter == "spring", ])$mean
t <- (mu_spring-mu_autumn)/SE

```

If the null hypothesis ($H_0 : \mu_{spring} = \mu_{autumn}$) were true, how likely is it to see a t-statistic this extreme (or more extreme) just by random chance? Now we find the p-value and evaluate our hypothesis:

```

p_value_2 <- 2 * (1 - pt(abs(t), df))
p_value_2

```

```
## [1] 0.005904168
```

Since $p < 0.05$, we reject H_0 , meaning that the difference in means is statistically significant.

3. Make a 95% confidence interval for the difference between mean wind speed in spring and autumn. What is the interpretation of the confidence interval?

The confidence interval is calculated as:

```

mu_conf <- mu_spring-mu_autumn
t_ask <- qt(1-0.05/2, df)
margin <- t_ask * SE

```

```
confidence_interval <- c(mu_conf-margin, mu_conf+margin)
confidence_interval
```

```
## [1] 0.07740868 0.42686643
```

We are 95% confident that the true difference in mean wind speed (spring autumn) lies between these two numbers and since the interval is above 0, spring wind speeds are significantly higher than autumn wind speeds.

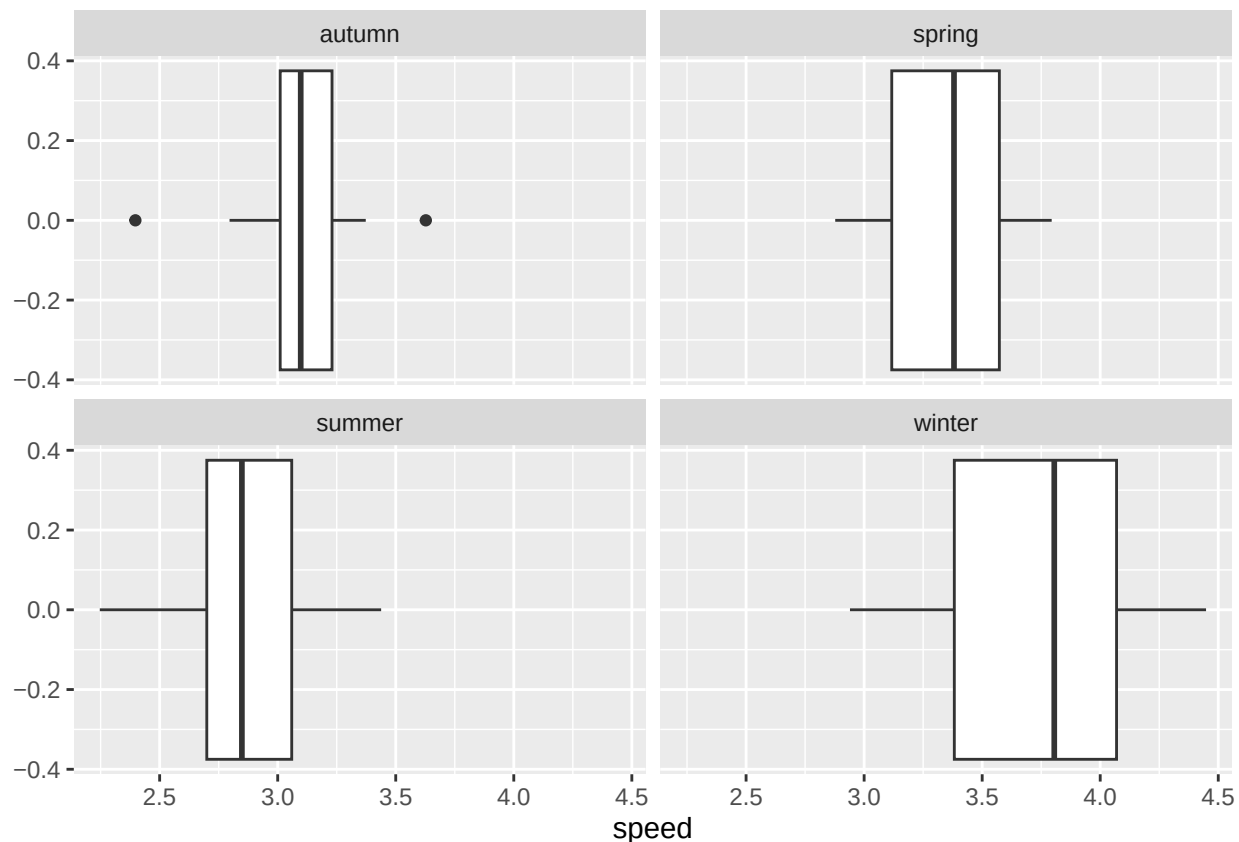
Part II: Comparing all four quarters

In this part we compare all four quarters of the year.

1. Make a box plot of the variable `speed` for each quarter and explain how a boxplot is computed.

A boxplot is obtained by sorting the data, then finding the quartiles (Q1, Q2, Q3). Next, a box is drawn in the interquartile range (Q3-Q1), and the median (Q2) is also marked. Finally, the whiskers are drawn as $Q1 - 1.5 \cdot IQR$ and $Q3 + 1.5 \cdot IQR$. Data points outside the IQR (outliers) are represented as dots.

```
gf_boxplot(~speed | quarter, data = wind)
```



2. Test the null-hypothesis that the mean wind speed is the same for all four quarters by editing the code below. Explain the parameter estimates in the output.

We are testing the null hypothesis

$$H_0 : \mu_{spring} = \mu_{summer} = \mu_{autumn} = \mu_{winter}$$

```
model <- lm(speed ~ quarter, data = wind)
summary(model)
```

```
##
## Call:
## lm(formula = speed ~ quarter, data = wind)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.78137 -0.18743  0.01865  0.20684  0.72687
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.09413    0.07105  43.551  < 2e-16 ***
## quarterspring  0.25214    0.10047   2.509   0.0143 *
## quartersummer -0.21538    0.10047  -2.144   0.0354 *
## quarterwinter  0.62729    0.10047   6.243 2.66e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3097 on 72 degrees of freedom
## Multiple R-squared:  0.5197, Adjusted R-squared:  0.4997
## F-statistic: 25.97 on 3 and 72 DF,  p-value: 1.711e-11
```

The Estimate column in the (Intercept) row represents the mean for autumn, and the rest of the values in Estimate are the difference of the rest of the quarters against autumn, in m/s. The differences in means are high (2.88m/s to 3.72m/s), and the value of Multiple R-squared: 0.5197 indicates that 52% of the total variability in wind speeds is explained by the quarter. Since the p-value of the F-test is less than 0.05 (1.711e-11), we reject the null hypothesis that the mean wind speed is the same for all four quarters.

To find out which quarters differ, we can make pairwise comparisons. We could use the pairwise t-tests as in Part I, Exercise 3. However, there is a problem with this approach:

3. How many pairwise comparisons of quarterly mean values can we make? What is the probability of rejecting a single true null-hypothesis? If we test several true null-hypotheses, explain why the probability of falsely rejecting at least one of them is larger than the significance level.

We can pair the quarters as spring vs summer, spring vs autumn, spring vs winter, summer vs autumn, summer vs winter, and autumn vs winter, so 6 pairs in total. A single true null hypothesis has a probability of being rejected of 0.05 (significance level $\alpha = 0.05$). Falsely rejecting would mean the opposite, so $P_{notreject} = 1 - \alpha = 0.95$. Falsely rejecting all 6 tests would be $0.95^6 = 0.735$. So the probability of a single false rejection is $1 - 0.735 = 0.265$. The more tests you perform, the higher the chance that at least one result looks significant purely by chance, since when many tests are performed, errors accumulate.

This phenomenon is known as multiple testing. We can adjust the pairwise t-tests for multiple testing by performing a so-called Tukey's test. It replaces the t-distribution by a different distribution (the studentized range distribution) that takes multiple testing into account.

4. Tukey's test can be performed using the code below. Which quarters differ significantly?

```
model<-aov(speed~quarter,data=wind)
TukeyHSD(model, conf.level=.95)
```

```
##      Tukey multiple comparisons of means
##      95% family-wise confidence level
```

```
##
## Fit: aov(formula = speed ~ quarter, data = wind)
##
## $quarter
##           diff          lwr          upr      p adj
## spring-autumn 0.2521376 -0.01211513 0.51639024 0.0668933
## summer-autumn -0.2153825 -0.47963516 0.04887022 0.1492955
## winter-autumn 0.6272860 0.36303327 0.89153864 0.0000002
## summer-spring -0.4675200 -0.73177271 -0.20326734 0.0000843
## winter-spring 0.3751484 0.11089571 0.63940109 0.0020779
## winter-summer 0.8426684 0.57841573 1.10692111 0.0000000
```

From Tukey's it can be understood that winter has a higher mean than all other quarters (all diff are high) and all p-values are very small. Summer and spring differ, with the latter showing significantly higher winds, and is statistically significant. The other two comparisons (spring-autumn and summer-autumn are not statistically significant since $p > 0.05$).

5. Compare the confidence interval for the difference between spring and autumn from Tukey's test to the one you computed in Part I, Exercise 3. What is different?

t-test $\rightarrow [0.07740868, 0.42686643]$

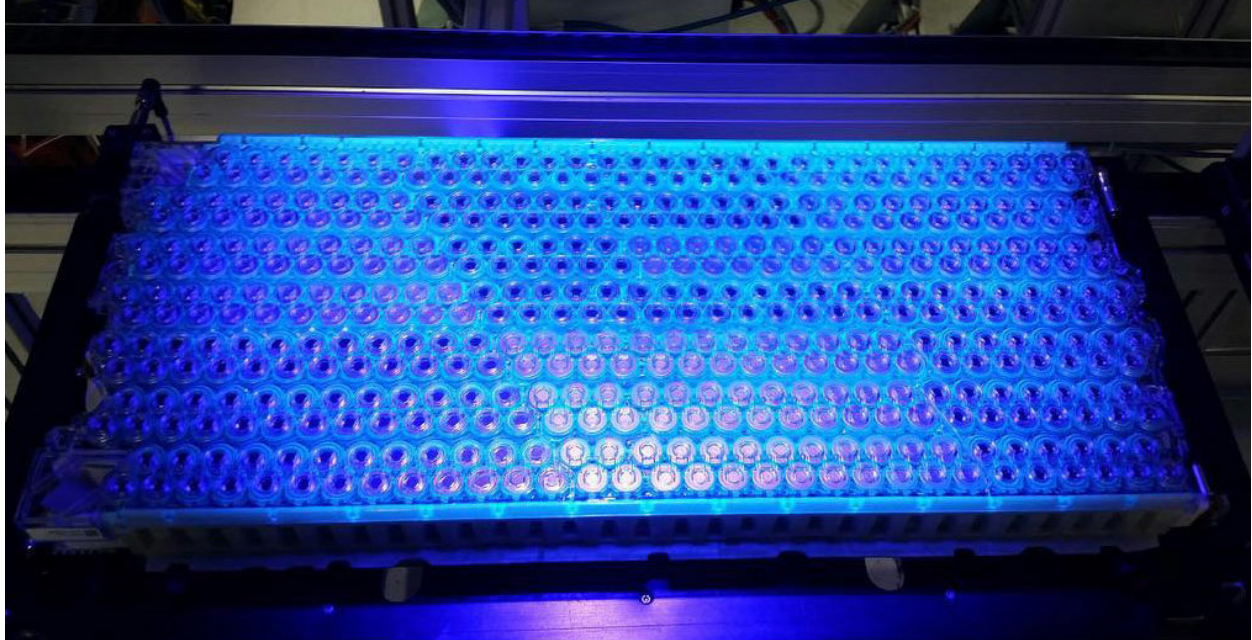
Does not include 0, hence it suggests a significant difference between spring and autumn.

Tukey's $\rightarrow [-0.01211513, 0.51639024]$

Wider, and includes 0, so spring and autumn are not significantly different after adjusting for multiple comparisons.

Battery capacity

Exam exercise: Battery capacity



In this exercise you will study a data set on battery capacity collected by researchers from AAU Energy. The batteries were kept at three different temperatures. The capacity loss Q (in percentage of initial capacity) is the response variable. We are interested in the effect on the response of the temperature and the efficient number of charge cycles FEC that the battery had been exposed to.

In the analysis, we will work with the log transformed variables $\log Q$ and $\log FEC$. Moreover, we will consider **Temperature** as a categorical variable with the three levels 35° , 40° , and 45° Celcius.

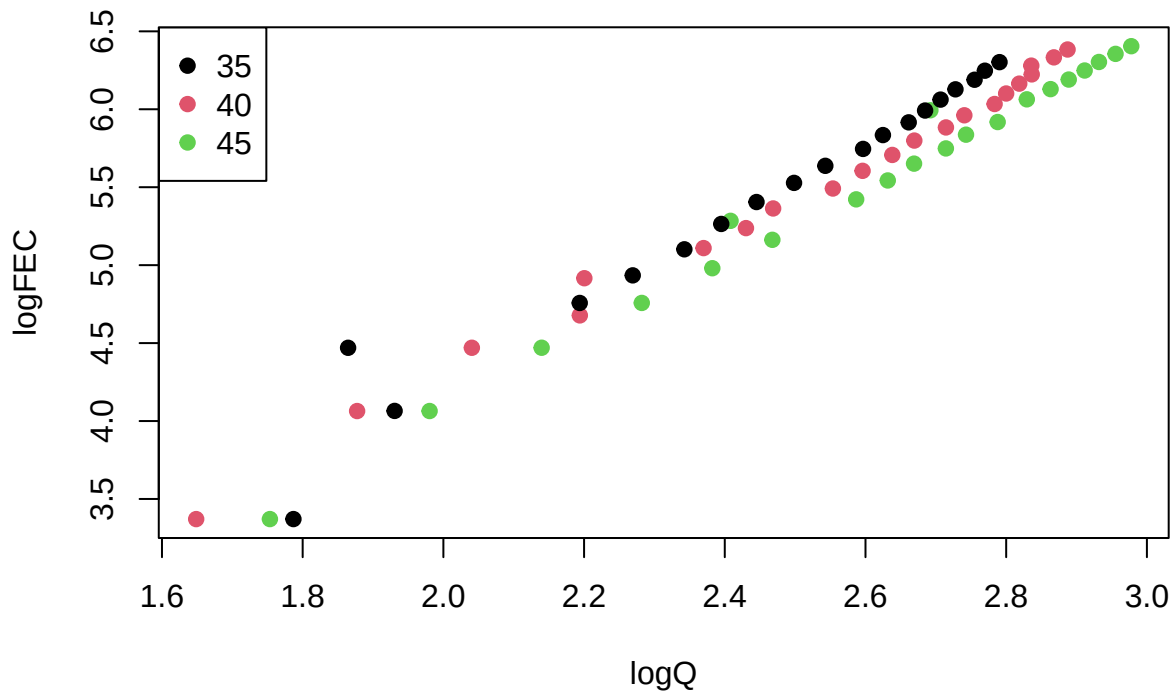
We begin by loading the **mosaic** package and reading in the data. Since we consider **Temperature** a categorical variable, we make this variable a factor.

```
library(mosaic)
capacity<-read.delim("https://asta.math.aau.dk/datasets?file=capacity.txt",sep="")
capacity$Temperature<-factor(capacity$Temperature)
```

1. Make a scatter plot of the log transformed variables $\log Q$ and $\log FEC$ where points are colored by temperature level.

```
plot(capacity$logQ, capacity$logFEC,
     col = capacity$Temperature, #categories
     pch = 19, #symbol for markers
     xlab = "logQ",
     ylab = "logFEC")
legend("topleft", legend = levels(capacity$Temperature),
```

```
col = 1:length(levels(capacity$Temperature)), pch = 19)
```



2. The code below finds the correlation between $\log Q$ and $\log \text{FEC}$ for each level of temperature. Explain the concept of correlation and interpret the results in relation to the plot from Question 1.

```
tempGrupper=split(capacity, capacity$Temperature)
lapply(tempGrupper, function(x) cor(x$logQ, x$logFEC))
```

```
## $`35`
## [1] 0.9813515
##
## $`40`
## [1] 0.9967649
##
## $`45`
## [1] 0.9943055
```

The absolute value of R (correlation) measures the (linear) strength of dependence between $\log Q$ and $\log \text{FEC}$. Since all three values are very close to 1, we can assume that $\log Q$ and $\log \text{FEC}$ are very correlated.

3. Consider a multiple regression model without interaction with $\log Q$ as the response variable and $\log \text{FEC}$ and Temperature as predictors. Write out the model equation using dummy variables. What is the interpretation of the parameters?

$$\log(Q) = \alpha + \beta_1 \cdot \log(\text{FEC}) + \beta_2 \cdot T + \epsilon$$

- α is the intercept. It corresponds to the value of $\log(Q)$ when both $\log(\text{FEC})$ and T are 0.
- β_1 and β_2 describe the change in the mean response ($\log(Q)$), when $\log(\text{FEC})$ or T are changed one unit respectively.
- ϵ is an error with a $\text{norm}(0, \sigma)$ distribution.

4. Edit the code below to fit the model from Question 3.


```
tempArray <- as.numeric(as.character(capacity$Temperature))
model<-lm(logQ ~ logFEC + tempArray,data=capacity)
summary(model)
```

```
##
## Call:
## lm(formula = logQ ~ logFEC + tempArray, data = capacity)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.193910 -0.010056  0.008889  0.017573  0.179922
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.152700   0.070099  -2.178   0.0335 *
## logFEC       0.410957   0.007488  54.885 < 2e-16 ***
## tempArray    0.010693   0.001461   7.317 8.58e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.04611 on 58 degrees of freedom
## Multiple R-squared:  0.9817, Adjusted R-squared:  0.9811
## F-statistic: 1558 on 2 and 58 DF,  p-value: < 2.2e-16
```

Explain the output from the code. Your explanation should as a minimum include:

- What is the prediction equation?

$$\log(Q) = \alpha + \beta_1 \cdot \log(\text{FEC}) + \beta_2 \cdot T + \epsilon$$

$$\log(Q) = -0.152700 + 0.410957 \cdot \log(\text{FEC}) + 0.010693 \cdot T + \epsilon$$

- Explain the calculation of *t* value and determination and interpretation of the p-value.

t value is calculated as $t = \frac{\text{Estimate}}{\text{Std. Error}}$.

The p-value is derived from the t-value using the t-distribution, which depends on the degrees of freedom. It quantifies the probability of getting a t-value that big (or bigger) if the true effect were actually zero.

- What is the interpretation of Multiple R-squared?

It measures the proportion of the variation in the response variable (*logQ*) that is explained by the predictors (*logFEC* and *Temperature*) in the model.

- Make an overall *F*-test for the null-hypothesis that there is no effect of any of the predictors.

```
anova(model)
```

```
## Analysis of Variance Table
##
## Response: logQ
##           Df Sum Sq Mean Sq  F value    Pr(>F)
## logFEC      1  6.5099   6.5099 3061.729 < 2.2e-16 ***
## tempArray   1  0.1138   0.1138  53.544 8.584e-10 ***
## Residuals  58  0.1233   0.0021
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

F value is the ratio of explained variance to unexplained variance and $\Pr(>F)$ is the p-value for the F-test of each predictor. From the values, we can reject the null hypothesis. At least one predictor (actually both in this case) significantly affects $\log(Q)$.

- Investigate whether or not there is an interaction between the effect of `Temperature` and the effect of `logFEC` as predictors of `logQ` (Hint: With more than two groups you need to use the `anova` function to make the test, see p. 31 in the slides from the second lecture of the module).

```
model_interaction <- lm(logQ ~ logFEC * tempArray, data=capacity)
summary(model_interaction)
```

```
##
## Call:
## lm(formula = logQ ~ logFEC * tempArray, data = capacity)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.207944 -0.003341  0.007317  0.017948  0.149931
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.459956   0.408299   1.127  0.264667
## logFEC         0.298821   0.074019   4.037  0.000163 ***
## tempArray     -0.004556   0.010119  -0.450  0.654228
## logFEC:tempArray 0.002789   0.001832   1.523  0.133386
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0456 on 57 degrees of freedom
## Multiple R-squared:  0.9824, Adjusted R-squared:  0.9815
## F-statistic: 1063 on 3 and 57 DF,  p-value: < 2.2e-16
```

```
anova(model, model_interaction)
```

```
## Analysis of Variance Table
##
## Model 1: logQ ~ logFEC + tempArray
## Model 2: logQ ~ logFEC * tempArray
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      58 0.12332
## 2      57 0.11850  1 0.0048198 2.3183 0.1334
```

Comparing the model without interaction to the model with interaction:

$F = 2.3183$, $p = 0.1334$, not significant.

Adding the interaction does not significantly improve the model. There is no significant interaction between `Temperature` and `logFEC`.

- Plot the model with and without interaction and explain the difference.

```
library(ggplot2)

# make sure Temperature is a factor
capacity$Temperature <- factor(capacity$Temperature)

# Fit models using the factor variable
model_no_interaction <- lm(logQ ~ logFEC + Temperature, data = capacity)
```

```
model_interaction <- lm(logQ ~ logFEC * Temperature, data = capacity)
```

```
summary(model_no_interaction)
```

```
##
## Call:
## lm(formula = logQ ~ logFEC + Temperature, data = capacity)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.197468 -0.004855  0.009251  0.017714  0.176427
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.224846   0.042060   5.346 1.65e-06 ***
## logFEC         0.411014   0.007511  54.720 < 2e-16 ***
## Temperature40  0.043310   0.014654   2.956  0.00453 **
## Temperature45  0.106577   0.014665   7.268 1.14e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.04625 on 57 degrees of freedom
## Multiple R-squared:  0.9819, Adjusted R-squared:  0.981
## F-statistic: 1032 on 3 and 57 DF, p-value: < 2.2e-16
```

```
summary(model_interaction)
```

```
##
## Call:
## lm(formula = logQ ~ logFEC * Temperature, data = capacity)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.222724 -0.001361  0.009642  0.015118  0.121917
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)     0.36911    0.07107   5.193  3.1e-06 ***
## logFEC           0.38439    0.01298  29.609 < 2e-16 ***
## Temperature40   -0.21810    0.09817  -2.222  0.03045 *
## Temperature45   -0.04996    0.09829  -0.508  0.61326
## logFEC:Temperature40  0.04798    0.01783   2.691  0.00942 **
## logFEC:Temperature45  0.02885    0.01780   1.621  0.11080
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.04425 on 55 degrees of freedom
## Multiple R-squared:  0.984, Adjusted R-squared:  0.9826
## F-statistic: 678.3 on 5 and 55 DF, p-value: < 2.2e-16
```

```
# Compare models (test for interaction)
```

```
anova(model_no_interaction, model_interaction)
```

```
## Analysis of Variance Table
```

```
##
```

```

## Model 1: logQ ~ logFEC + Temperature
## Model 2: logQ ~ logFEC * Temperature
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      57 0.12195
## 2      55 0.10767  2  0.014278 3.6465 0.03257 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# Create grid for predictions using Temperature levels (same names as model)
newdat <- expand.grid(
  logFEC = seq(min(capacity$logFEC, na.rm = TRUE),
               max(capacity$logFEC, na.rm = TRUE),
               length.out = 200),
  Temperature = levels(capacity$Temperature)
)

# Predict from both models (now newdat matches the model variables)
newdat$pred_no_interaction <- predict(model_no_interaction, newdata = newdat)
newdat$pred_interaction    <- predict(model_interaction, newdata = newdat)

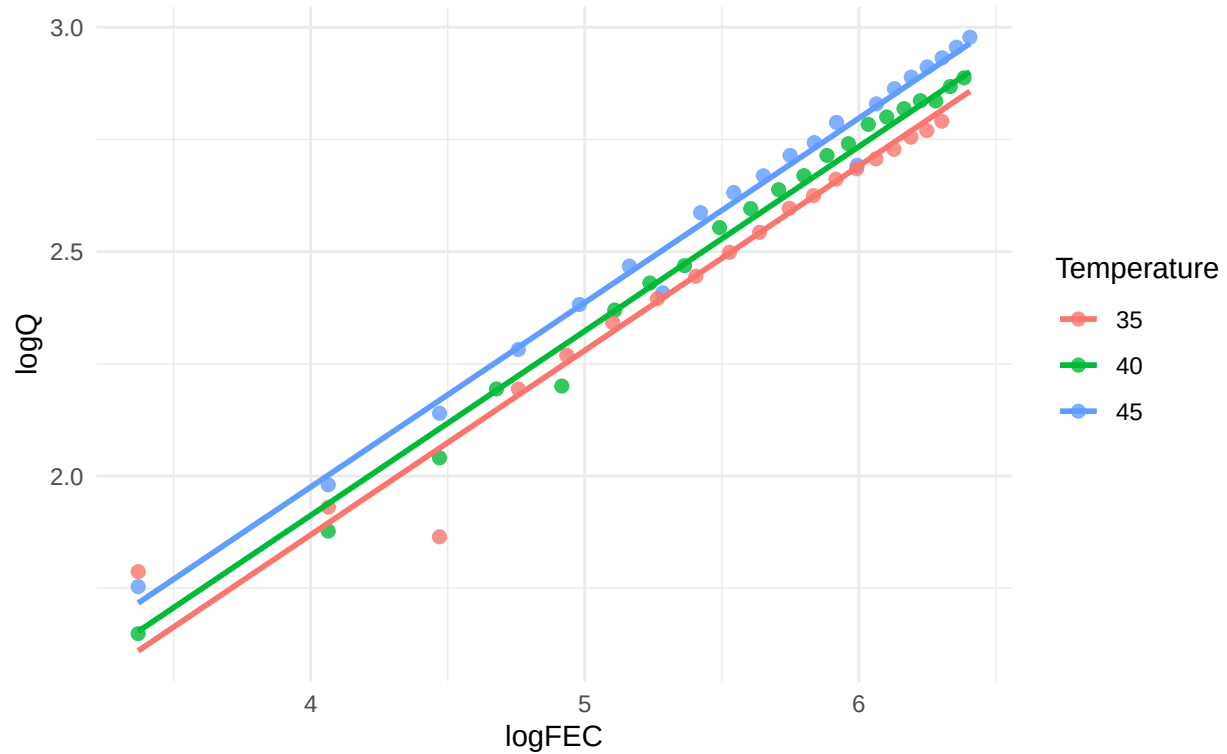
ggplot(capacity, aes(x = logFEC, y = logQ, color = Temperature)) +
  geom_point(alpha = 0.8, size = 2) +
  geom_line(
    data = newdat,
    aes(y = pred_no_interaction),
    size = 1
  ) +
  labs(
    title = "No-interaction model",
    subtitle = "Common slope across temperatures (parallel lines)",
    x = "logFEC",
    y = "logQ"
  ) +
  theme_minimal()

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

No-interaction model

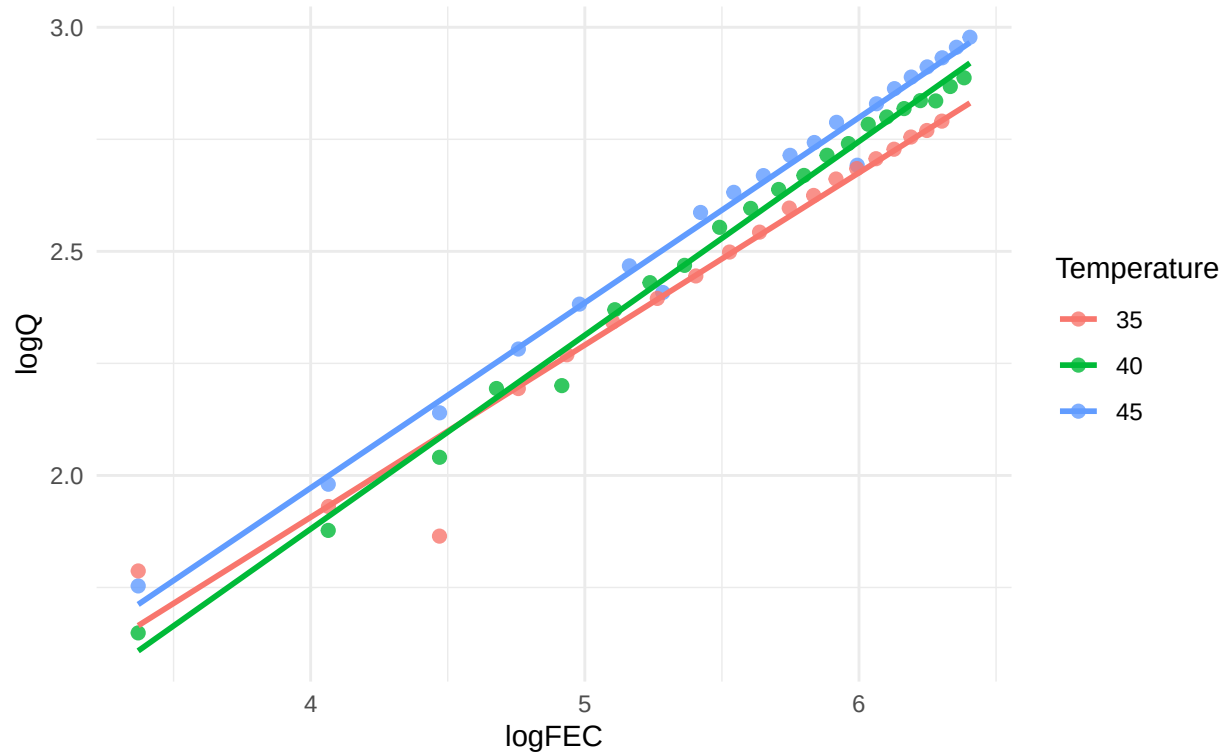
Common slope across temperatures (parallel lines)



```
ggplot(capacity, aes(x = logFEC, y = logQ, color = Temperature)) +  
  geom_point(alpha = 0.8, size = 2) +  
  geom_line(  
    data = newdat,  
    aes(y = pred_interaction),  
    size = 1  
  ) +  
  labs(  
    title = "Interaction model",  
    subtitle = "Different slopes by temperature",  
    x = "logFEC",  
    y = "logQ"  
  ) +  
  theme_minimal()
```

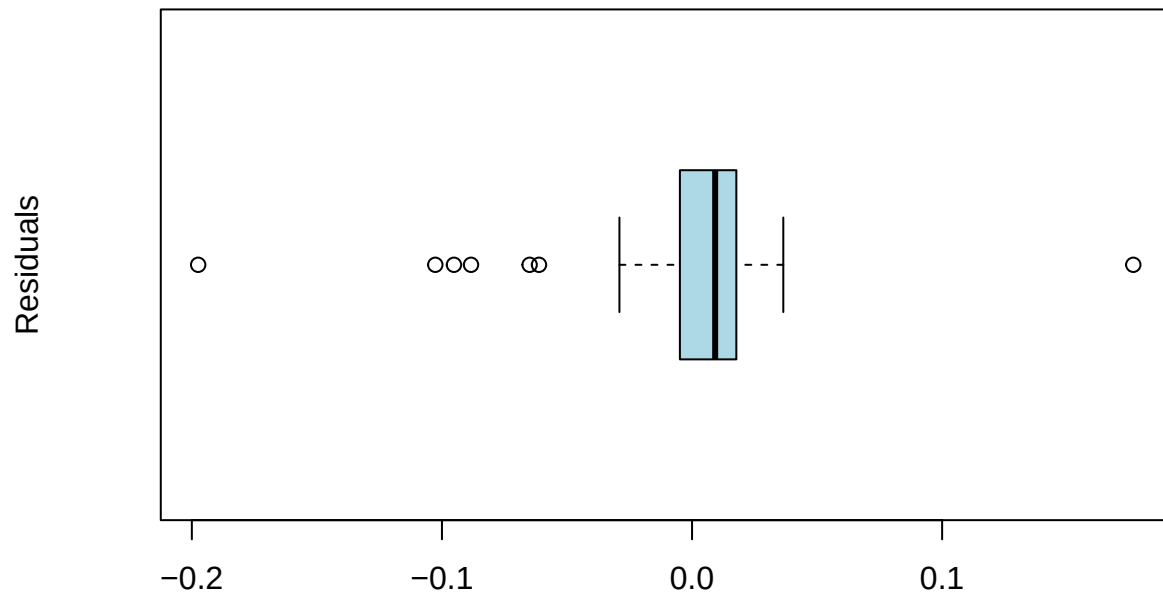
Interaction model

Different slopes by temperature



```
# Residual boxplot for the no-interaction model
residuals_no_int <- residuals(model_no_interaction)
boxplot(residuals_no_int,
        main = "Boxplot of residuals (no-interaction model)",
        ylab = "Residuals",
        col = "lightblue", horizontal = TRUE)
abline(h = 0, col = "red", lty = 2)
```

Boxplot of residuals (no-interaction model)



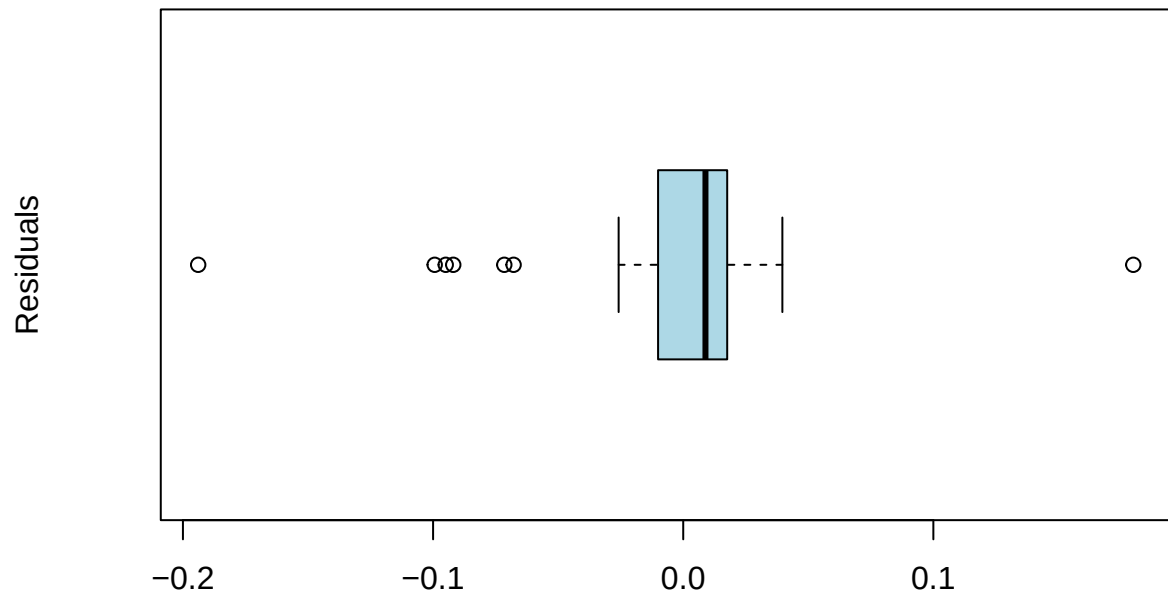
7. The code below computes the residuals $y_i - \hat{y}_i$ in the model without interaction. Draw a boxplot of the residuals and comment on the result.

```
# compute residuals
residuals <- model$residuals # or m0$residuals if using the variable from previous chunk

# create a boxplot of residuals
boxplot(residuals,
  main = "Boxplot of residuals (no-interaction model)",
  ylab = "Residuals",
  col = "lightblue",
  horizontal = TRUE)

# optionally, add a horizontal line at 0
abline(h = 0, col = "red", lty = 2)
```

Boxplot of residuals (no-interaction model)



The median residual (thick line in the box) is very close to 0, which suggests that the model predictions are roughly unbiased on average. There are several points outside the whiskers, on both the left and right sides. Negative outliers (~ -0.2 to -0.1) indicate that the model overestimates $\log Q$ for these observations.

Exam exercise for Module 4: A model for a DC-motor



The purpose of this exam question is to provide a model for how the velocity of a DC-motor depends on the input voltage. For this, data was collected from an experiment. Input-output data are recorded in the data set below. The first column is the input voltage (V) and the second column is the motor velocity (rad/s). The sampling time is 0.01 s. A linear regression with ARMA noise should be obtained.

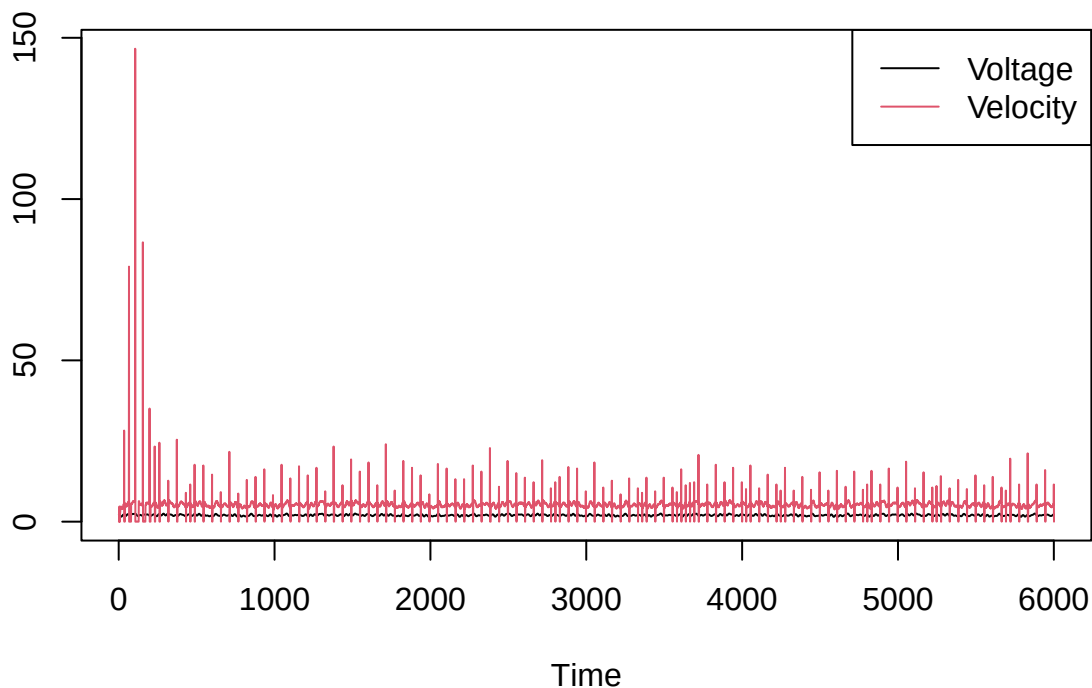
Viewing and cleaning data

- Load the data into R:

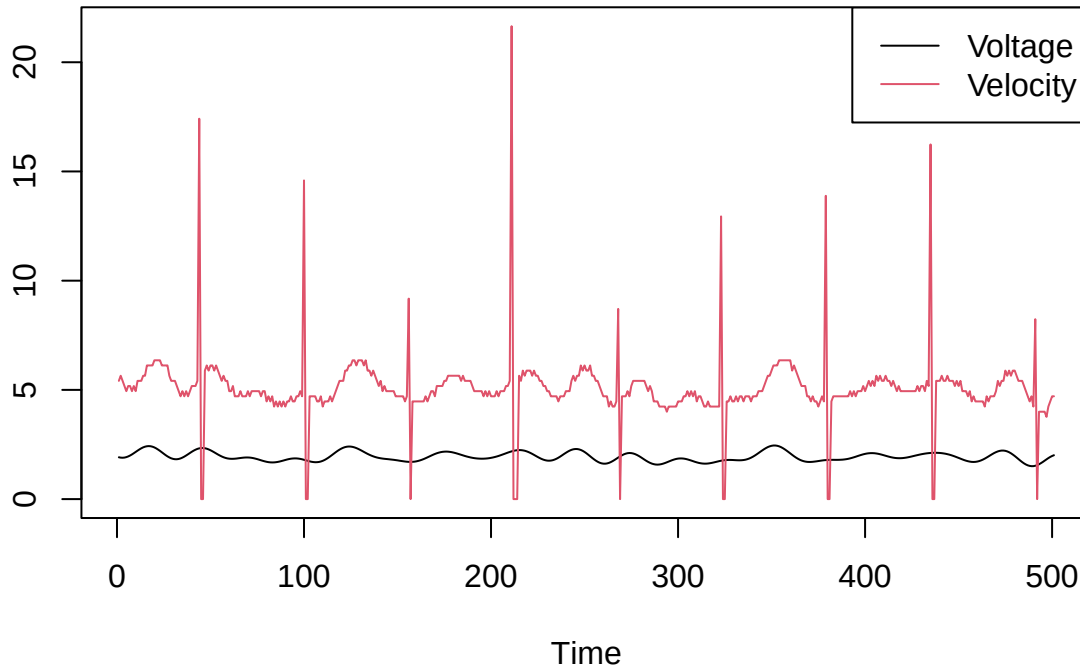
```
dat = as.ts(read.table("https://asta.math.aau.dk/datasets?file=dcmotor.txt"),
             colnames=c("Voltage", "Velocity"))
```

1. Start by plotting the data (both input and output). You may want to zoom in on a part to get a better view.

```
ts.plot(dat,col=1:2); legend("topright",legend=c("Voltage", "Velocity"),col=1:2,lty=1)
```

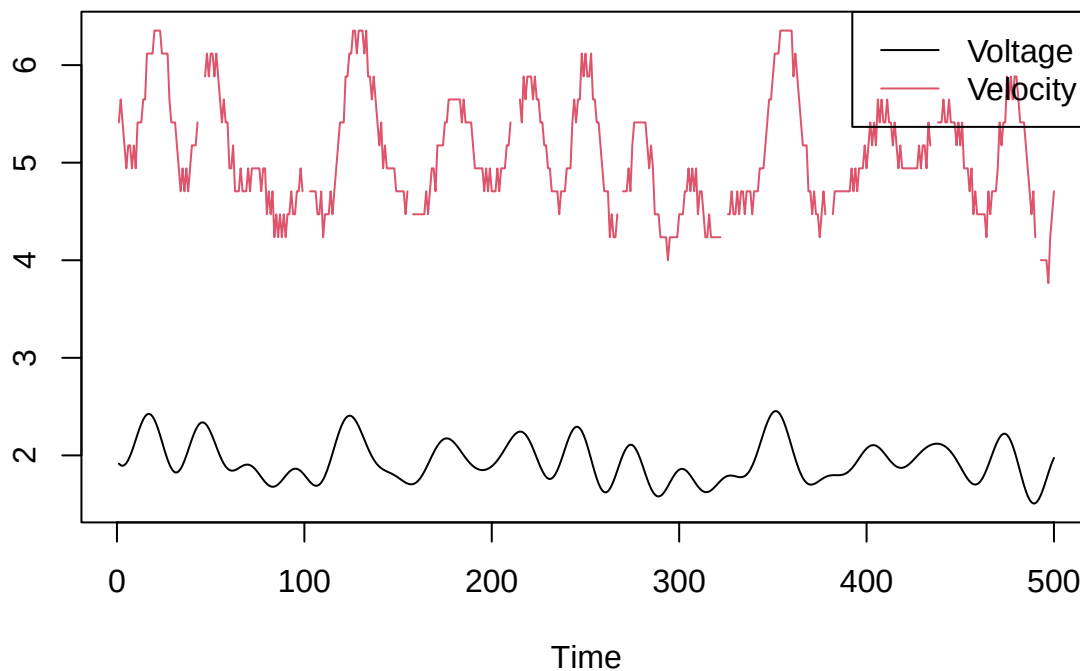


```
ts.plot(dat[500:1000,],col=1:2); legend("topright",legend=c("Voltage","Velocity"),col=1:2,lty=1)
```



2. It should be obvious that the output data is subject to noise (especially in the beginning) and a regularly repeating unknown disturbance. Choose a part of the data that sounds reasonable for obtaining the model. For example, here is a part of the data with the first 500 measurements removed, and all measurements outside the interval $[3, 7]$ removed.

```
datX = dat[500:6001,] # removing start
datX[datX[,2]<3|datX[,2]>7,2]=NA # exchanging velocity outliers outside interval [3,7] with NA
ts.plot(datX[1:500,],col=1:2); legend("topright",legend=c("Voltage","Velocity"),col=1:2,lty=1)
```



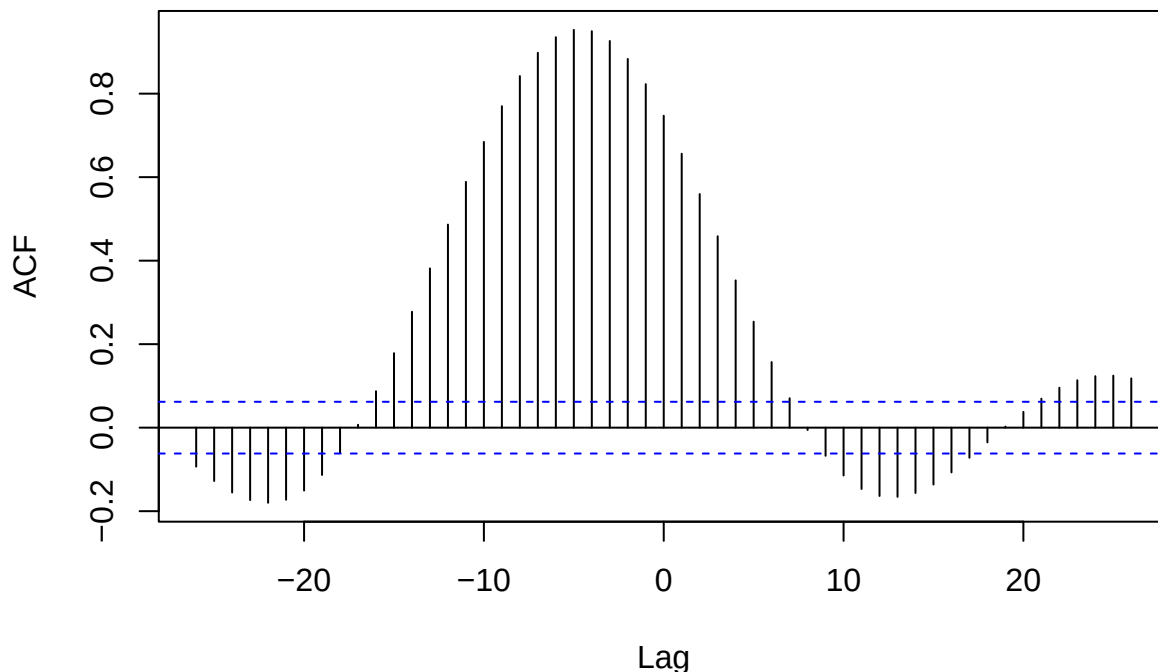
```
dat1 = datX[1000:2000,] # cherry-picking a time interval
```

A linear regression model with ARMA noise

3. Use the cross-correlation function to detect the delay between the input and output (NA values can be handled in the ccf as well as the acf by using the option na.action=na.pass).
 - Explain the cross-correlation function.

```
cc = ccf(dat1[, "Voltage"], dat1[, "Velocity"], na.action=na.pass)
```

dat1[, "Voltage"] & dat1[, "Velocity"]



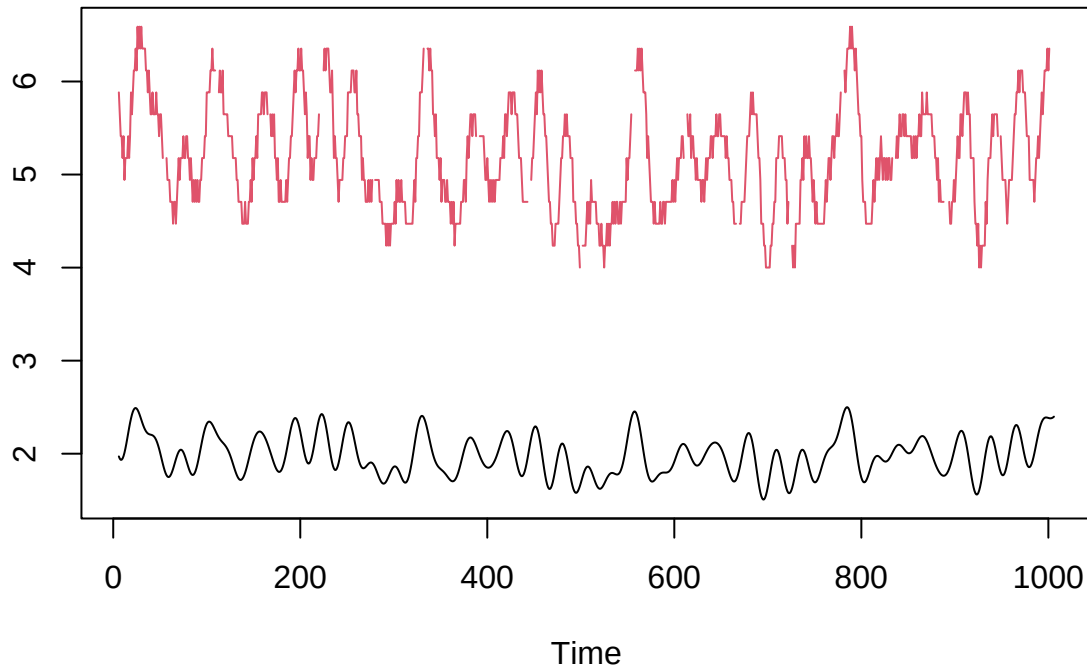
```
estlag = cc$lag[which(cc$acf==max(abs(cc$acf)))]
estlag # in samples, but since sampling time is 0.01s, 1 sample = 10ms
```

```
## [1] -5
```

The cross-correlation function describes the linear dependence between two time series as a function of a time lag. Peaks in the CCF indicate delays where the input has the strongest influence on the output. The lag corresponding to the maximum absolute correlation provides an estimate of the system delay, which is required when formulating the regression model. There is a system delay of -40ms.

4. Fit a linear regression model with AR(1) noise for the output using the input as exogenous variable (adjust for the delay using the lag() function in R). What is the autocorrelation function for the noise process in this model?

```
dat_shifted = ts.intersect(lag(as.ts(dat1[, "Voltage"]),estlag),dat1[, "Velocity" ])
colnames(dat_shifted) <- c("Voltage", "Velocity")
ts.plot(dat_shifted[, "Voltage"],dat_shifted[, "Velocity"],col=1:2)
```



```
mod=arima(dat_shifted[, "Velocity"],order=c(1,0,0),xreg=dat_shifted[, "Voltage"]); mod
```

```
##
## Call:
## arima(x = dat_shifted[, "Velocity"], order = c(1, 0, 0), xreg = dat_shifted[,
##      "Voltage"])
##
## Coefficients:
##          ar1  intercept  dat_shifted[, "Voltage"]
##      0.8584      1.6348          1.7945
## s.e.  0.0168      0.2896          0.1442
##
## sigma^2 estimated as 0.0355:  log likelihood = 227.55,  aic = -447.1
```

This results in the model:

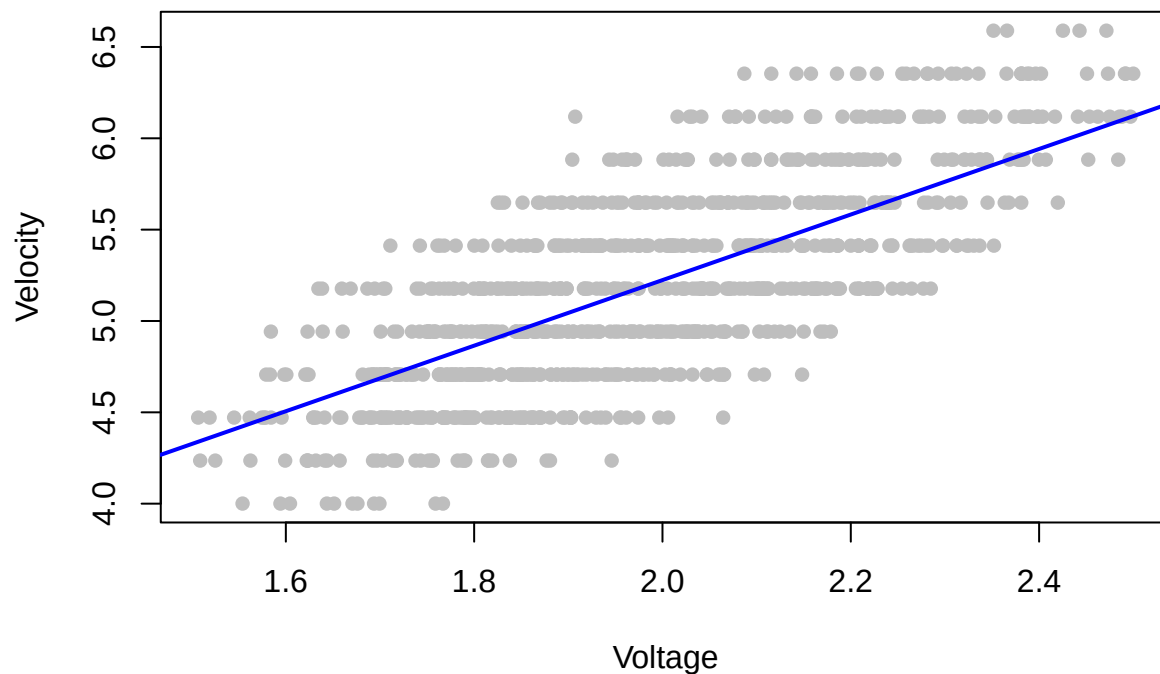
$$\omega_t = \beta_0 + \beta_1 \cdot V_t + \alpha \cdot \epsilon_{t-1} + W_t = 1.6348 + 1.7945 \cdot V_t + 0.8584 \cdot \epsilon_{t-1} + W_t$$

The autocorrelation function measures how correlated X_t and X_{t+h} are related. Assuming an AR(1) and stationary process (each value depends on previous value + noise, and the mean and variance are constant):

$$\rho(t, t+h) = \alpha^h = 0.8584^h$$

Plotting the model:

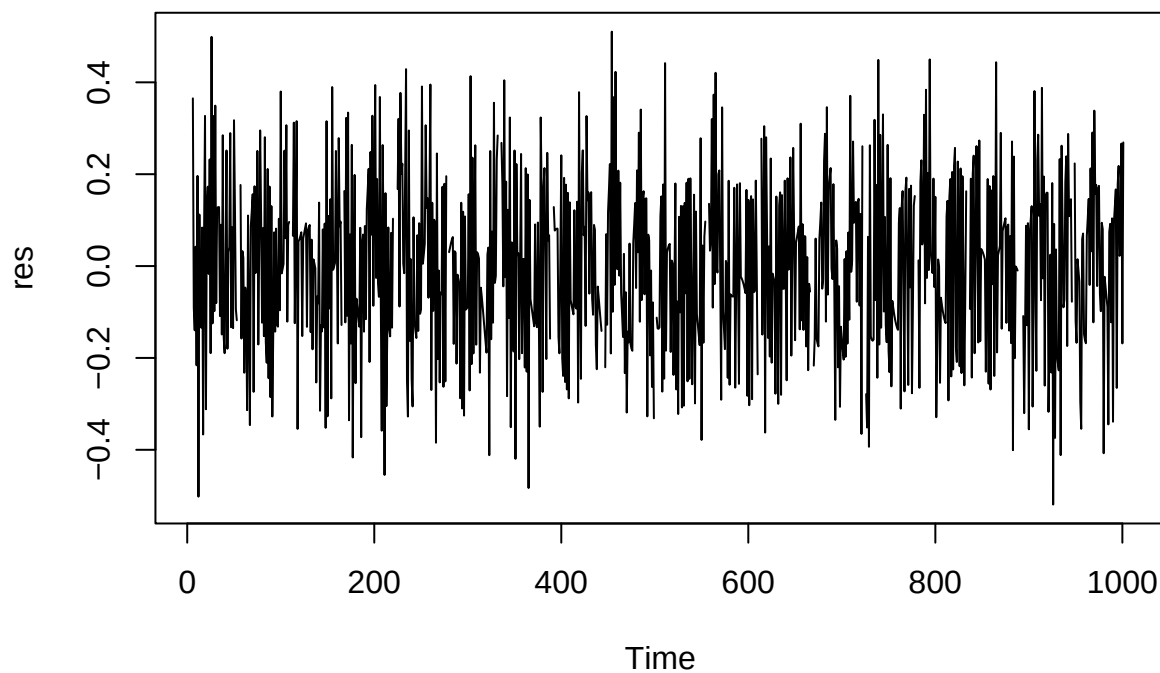
```
# Scatter plot of Voltage vs Velocity
plot(dat_shifted[, "Voltage"], dat_shifted[, "Velocity"], pch = 16, col = "grey", xlab = "Voltage", ylab = "Velocity")
abline(a = 1.6348, b = 1.7945, col = "blue", lwd = 2)
```



5.

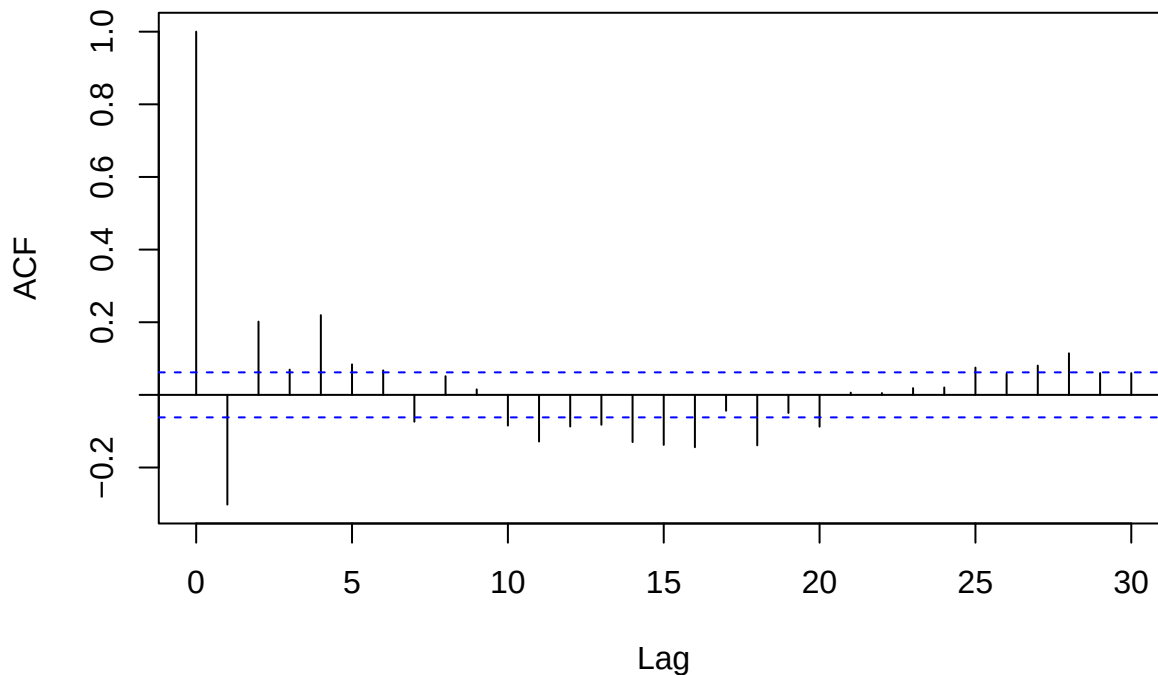
Check whether the AR(1) model fits the data by considering the residuals. - Use a time series plot of the residuals as well as a correlogram. - Explain what you expect to see if your model is correct.

```
res=resid(mod)
plot(res)
```



```
acf(res, na.action=na.pass)
```

Series res



The residuals should look like white noise in both plots. As expected, The time series looks like a random distribution with $\mu = 0$ and the correlogram fits the expected shape for white noise, although there remains some of the oscillations which are meaningful from the original data.

6. Fit a linear regression model with ARMA(p,q) noise for different choices of p and q. Compare the models using AIC and pick the preferred model (if you get a warning message, try using the option `optim.control=list(maxit = 1000)`).

```
# List of ARMA(p,q) orders you want to try
# All combinations of p and q from 0 to 3
p_values <- 0:3
q_values <- 0:3

# Data frame to store results
results <- data.frame(p=integer(),
                      q=integer(),
                      AIC=numeric(),
                      stringsAsFactors=FALSE)

# Loop over all combinations
for (p in p_values) {
  for (q in q_values) {
    fit <- try(arima(dat_shifted[, "Velocity"],
                    order = c(p,0,q),
                    xreg = dat_shifted[, "Voltage"],
                    optim.control = list(maxit=1000)),
              silent = TRUE)

    if (!inherits(fit, "try-error")) {
      cat("Fitted ARMA(", p, ", 0, ", q, ")\\n")
    }
  }
}
```

```

    results <- rbind(results, data.frame(p=p, q=q, AIC=fit$aic))
  } else {
    cat("ARMA(", p, ",0,", q, ") failed to converge\n")
  }
}
}

```

```

## Fitted ARMA( 0 ,0, 0 )
## Fitted ARMA( 0 ,0, 1 )
## Fitted ARMA( 0 ,0, 2 )
## Fitted ARMA( 0 ,0, 3 )
## Fitted ARMA( 1 ,0, 0 )
## Fitted ARMA( 1 ,0, 1 )
## Fitted ARMA( 1 ,0, 2 )
## Fitted ARMA( 1 ,0, 3 )
## Fitted ARMA( 2 ,0, 0 )
## Fitted ARMA( 2 ,0, 1 )
## Fitted ARMA( 2 ,0, 2 )

```

```

## Warning in arima(dat_shifted[, "Velocity"], order = c(p, 0, q), xreg =
## dat_shifted[, : possible convergence problem: optim gave code = 1

```

```

## Fitted ARMA( 2 ,0, 3 )
## Fitted ARMA( 3 ,0, 0 )
## Fitted ARMA( 3 ,0, 1 )
## Fitted ARMA( 3 ,0, 2 )
## Fitted ARMA( 3 ,0, 3 )

```

```

print(results)

```

```

##      p q      AIC
## 1  0 0  780.72437
## 2  0 1  247.31250
## 3  0 2  -78.74443
## 4  0 3 -208.93120
## 5  1 0 -447.09979
## 6  1 1 -537.02235
## 7  1 2 -596.81330
## 8  1 3 -713.84759
## 9  2 0 -569.87636
## 10 2 1 -568.38824
## 11 2 2 -565.91753
## 12 2 3 -970.77009
## 13 3 0 -568.69875
## 14 3 1 -567.06614
## 15 3 2 -564.39158
## 16 3 3 -980.50312

```

```

# Select the best model

```

```

best_model <- results[which.min(results$AIC), ]
cat("Best model is ARMA(", best_model$p, ",0,", best_model$q, ") with AIC =", best_model$AIC, "\n")

```

```

## Best model is ARMA( 3 ,0, 3 ) with AIC = -980.5031

```

Apparently, the higher the order, the lower the AIC. We choose the maximum suggested value for p and q of 3.

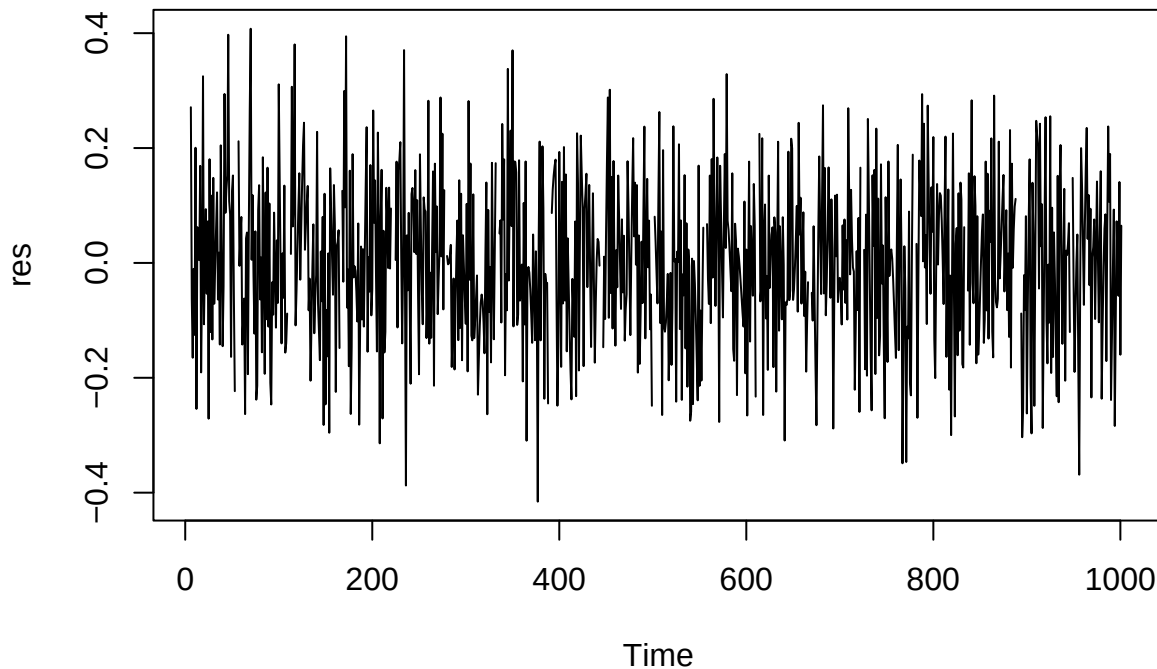
7. Check whether the chosen model fits the data by considering the correlogram for the residuals.

- If the model does not fit well you may consider returning to the previous step and increase the order of the ARMA(p,q) model up to $p \leq 3$ and $q \leq 3$ to see if you can get a model that fits better.

```
mod=arima(dat_shifted[, "Velocity"],order=c(3,0,3),xreg=dat_shifted[, "Voltage"], optim.control = list(maxit=1000))
mod
```

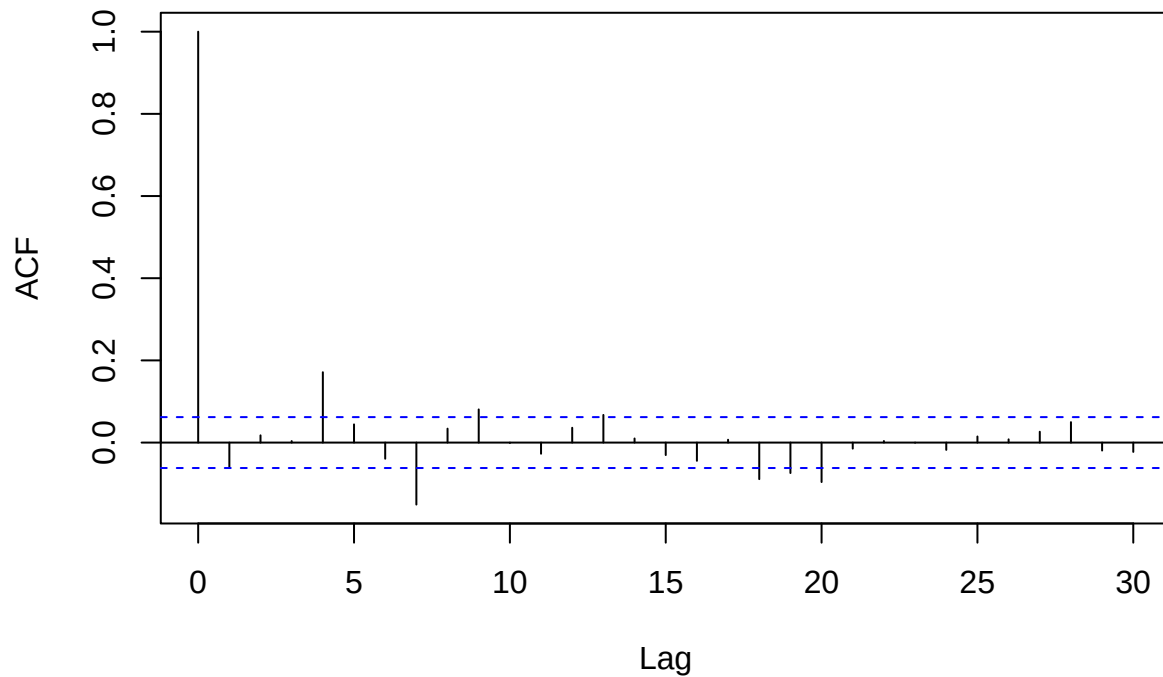
```
##
## Call:
## arima(x = dat_shifted[, "Velocity"], order = c(3, 0, 3), xreg = dat_shifted[,
##      "Voltage"], optim.control = list(maxit = 1000))
##
## Coefficients:
## Warning in sqrt(diag(x$var.coef)): NaNs produced
##      ar1      ar2      ar3      ma1      ma2      ma3  intercept
##      0.9241  0.955 -0.9682 -0.7027 -0.8287  0.8740    0.4916
## s.e.  0.0088   NaN   0.0088  0.0217  0.0097  0.0169    0.2225
##      dat_shifted[, "Voltage"]
##              2.3687
## s.e.              0.1118
##
## sigma^2 estimated as 0.0199:  log likelihood = 499.25,  aic = -980.5
```

```
res=resid(mod)
plot(res)
```



```
acf(res, na.action=na.pass)
```


Series res



8. When you have a model that fits the data, write out the model and interpret the estimated parameters.

The selected model is a linear regression with ARMA(3,3) errors:

$$\omega_t = \beta_0 + \beta_1 \cdot V_t + n_t, n_t = \sum_{i=1}^3 \phi_i n_{t-i} + \sum_{j=1}^3 \theta_j \varepsilon_{t-j} + \varepsilon_t$$

$$\omega_t = 0.4916 + 2.3687 V_t + n_t,$$

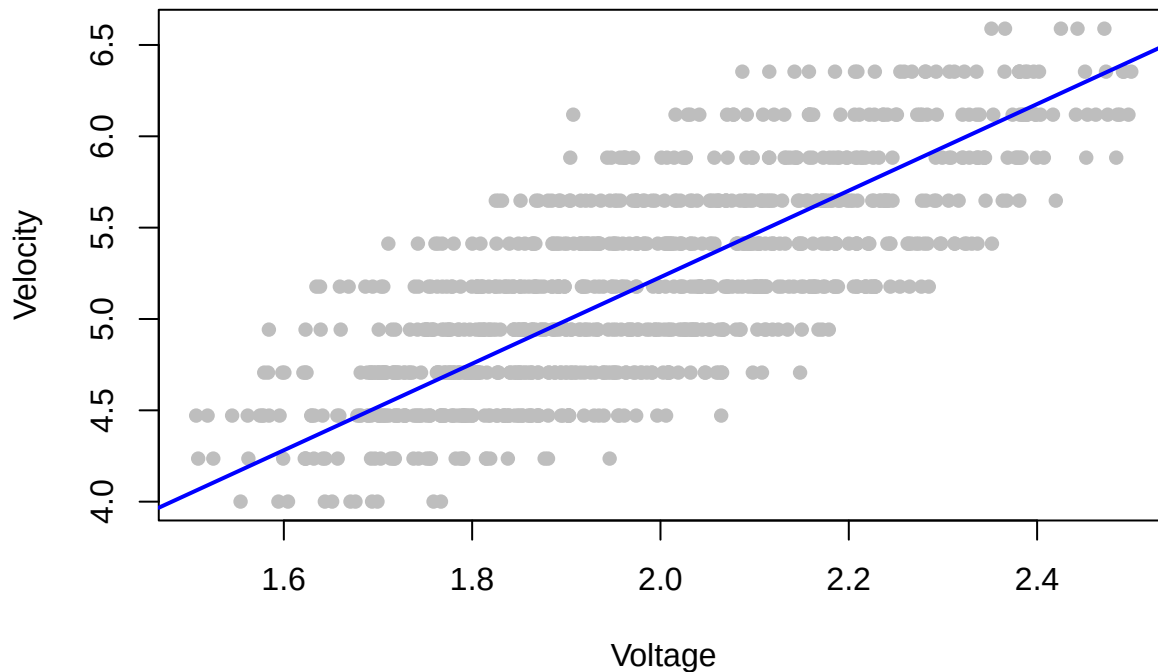
$$n_t = 0.9241 n_{t-1} + 0.9550 n_{t-2} - 0.9682 n_{t-3}$$

$$+ \varepsilon_t - 0.7027 \varepsilon_{t-1} - 0.8287 \varepsilon_{t-2} + 0.8740 \varepsilon_{t-3}$$

```
# Scatter plot of Voltage vs Velocity
```

```
plot(dat_shifted[, "Voltage"], dat_shifted[, "Velocity"], pch = 16, col = "grey", xlab = "Voltage", ylab = "Velocity")
```

```
abline(a = 0.4916, b = 2.3687, col = "blue", lwd = 2)
```



9. Make a confidence interval for the slope of the regression.

```
slope <- mod$coef["dat_shifted[, \"Voltage\"]"]

# Extract the standard error
se_slope <- sqrt(mod$var.coef["dat_shifted[, \"Voltage\"]", "dat_shifted[, \"Voltage\"]"])

# Calculate 95% Confidence Interval
lower_ci <- slope - 2 * se_slope
upper_ci <- slope + 2 * se_slope

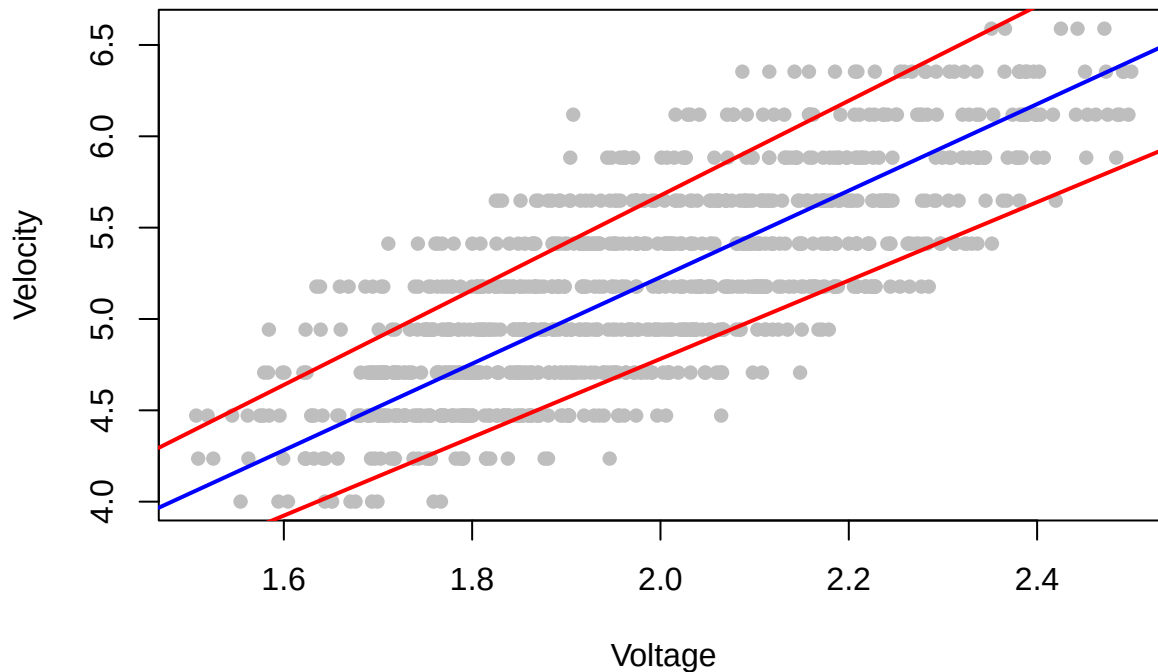
cat("Slope Estimate:", slope, "\n")

## Slope Estimate: 2.368654

cat("95% CI: [", lower_ci, ",", upper_ci, "]\n")

## 95% CI: [ 2.145045 , 2.592264 ]

plot(dat_shifted[, "Voltage"], dat_shifted[, "Velocity"], pch = 16, col = "grey", xlab = "Voltage", ylab = "Velocity")
abline(a = 0.4916, b = 2.3687, col = "blue", lwd = 2)
abline(a = 0.4916, b = lower_ci, col = "red", lwd = 2)
abline(a = 0.4916, b = upper_ci, col = "red", lwd = 2)
```



10.

Suppose the next value of the voltage is 2.23. Predict the next observation of velocity and make a confidence interval for the prediction.

```
# New data for prediction
new_voltage <- 2.23
new_xreg <- matrix(new_voltage, nrow = 1)

# Predict the next value using the fitted model
pred <- predict(mod, n.ahead = 1, newxreg = new_xreg)
pred

## $pred
## Time Series:
## Start = 1007
## End = 1007
## Frequency = 1
## [1] 5.85348
##
## $se
## Time Series:
## Start = 1007
## End = 1007
## Frequency = 1
## [1] 0.1571207

predicted_velocity <- pred$pred

cat("Predicted velocity at ", new_voltage, " V:", predicted_velocity, " rad/s\n")

## Predicted velocity at  2.23  V: 5.85348  rad/s

se_pred <- pred$se

# Calculate the 95% prediction interval
lower_pred_ci <- predicted_velocity - 2 * se_pred
```

```
upper_pred_ci <- predicted_velocity + 2 * se_pred
cat("95% Prediction Interval: [", lower_pred_ci, ",", upper_pred_ci, "]\n")

## 95% Prediction Interval: [ 5.539239 , 6.167722 ]
```